Student Skill Exchange

Pre-Defense Assessment Report

Diploma topic	Development of a Peer-to-Peer Skill Exchange Platform for University Students
Team	Makhmutov Nursultan; Burlibaev Tamir (SE-2413)
Supervisor	Latuta Konstantin
Institution	Astana IT University
Prepared	23 October 2026
Repository	https://github.com/HashiraOfSun/student-skill-exchange
Project board	https://student-skill-exchange.youtrack.cloud/agiles/204-1/current

Status note: this report consolidates the approved project documentation and interactive UI prototype. The repository does not yet contain a runnable production application. Verification and evaluation activities are therefore presented as planned work; no implementation or test results are claimed.

Report structure

1. Project proposal and scope

2. Relevance evidence and student research

3. Literature review

4. Existing systems comparison

5. Technology comparison and selected stack

6. Requirements

7. System architecture

8. Interactive UI prototype

9. Verification and evaluation plan

10. Project management, responsibilities, and risks

11. Supervisor feedback and revisions

Document basis and limitations

- All narrative sections are compiled from files merged into the repository's main branch.
- The student research record documents the supplied research process and its limitations.
- Product comparisons rely primarily on official product sources and clearly label unverified capabilities.
- The UI deliverable is an interactive HTML prototype; it is not evidence of a deployed application.
- The verification plan distinguishes planned tests from tests already performed.

1. Project proposal and scope

Source document: docs/project-proposal.md

Project title

Development of a Peer-to-Peer Skill Exchange Platform for University Students.

Team and supervisor

- **Nursultan Makhmutov**, SE-2413 - backend development.
- **Tamir Burlibaev**, SE-2413 - frontend development.
- **Supervisor:** Konstantin Latuta.
- **University:** Astana IT University.

Problem statement

Students have different practical skills and learning interests.
A student who can help with programming may need help practising
English, while another student may have the opposite strengths.

Our starting assumption is that finding a suitable exchange partner
through general student chats can take time. Information about what
people can teach and what they want to learn may be scattered across
messages, making reciprocal matches difficult to identify.

We will investigate this assumption through discussions with 3-5
potential users. At this stage, we have not yet established how
common this problem is or whether students would regularly use
a dedicated platform.

Target users

The initial target users are university students who want to share their knowledge and learn practical skills from other students.

We will initially focus our user research on students at Astana IT University. The first version will support individual exchanges between two participants.

Project aim

To develop and evaluate a web platform that helps university students find partners with complementary skills and arrange peer-to-peer learning exchanges.

Objectives

1. Investigate students' current learning support practices, difficulties, and expectations.
2. Review relevant research and compare existing platforms.
3. Define functional and non-functional requirements with verifiable acceptance criteria.
4. Compare technology options and design the system architecture.
5. Develop user accounts, profiles, and skill management.
6. Implement search and reciprocal partner matching.
7. Implement exchange requests and agreement on exchange details.
8. Integrate the frontend and backend.
9. Test the main scenarios and evaluate the matching approach against a defined baseline.
10. Prepare technical documentation and user instructions.

Proposed solution

The platform will allow each student to list skills they can offer and skills they want to learn. Students will be able to search for participants and view suggested exchange partners.

The initial matching approach will look for reciprocal matches: student A offers a skill wanted by student B, and student B offers a skill wanted by student A. Recommendations will show the skills that produced the match.

A student can send an exchange request to another participant. The recipient can accept or decline it, and requests can be cancelled according to rules defined in the requirements.

Participants will be able to record the agreed topic, time, and format of the exchange. The learning session itself will take place outside the platform, either in person or through an external communication service.

Main user scenario

1. A student creates an account and signs in.
2. The student adds offered and desired skills to their profile.
3. The student searches for a partner or views recommendations.
4. The student opens a profile and sends an exchange request.
5. The recipient reviews and accepts or declines the request.
6. If accepted, the participants agree on the topic, time, and format of the exchange.
7. Both participants can view the recorded exchange details.

Scope of the first version

The first version will include:

- Registration, login, and access control.
- User profiles and profile editing.
- Offered and desired skills.
- Search and filtering by skills and categories.
- Partner recommendations based on reciprocal skill matches.
- An explanation of matching skills.
- Incoming and outgoing exchange requests.
- Request statuses and status history.
- Recording agreed exchange details.
- A web interface adapted for desktop and mobile screens.

The first version will not include:

- Payments or paid tutoring.
- Built-in video calls.
- A real-time chat system.
- Native mobile applications.
- Machine-learning-based recommendations.
- Certificates or formal assessment of users' skills.
- Group classes or a course marketplace.

Engineering challenges

The project requires consistent interaction between user profiles, skill data, partner matching, and exchange requests.

The backend must prevent unauthorized changes to other users' profiles and requests. Request status transitions must follow

defined rules, and both participants must see consistent information.

The matching logic must handle incomplete profiles, repeated skills, and cases where no reciprocal partner is available. The frontend must communicate these situations clearly.

Responsibilities

Nursultan Makhmutov

- Design the backend architecture and database.
- Develop the REST API.
- Implement authentication, validation, and access control.
- Implement profiles, skills, search, and partner matching.
- Implement exchange request processing and data storage.
- Test backend functionality and document the API.

Tamir Burlibaev

- Design user flows, page structure, and interface mockups.
- Develop registration, login, and profile interfaces.
- Build search and recommendation screens.
- Develop exchange request and agreement interfaces.
- Integrate the frontend with the API.
- Handle validation messages, loading states, and errors.
- Check responsive layouts and navigation.

Shared responsibilities

Both team members will contribute to research, requirements, technology decisions, integration, testing, and reporting. Individual tasks and contributions will be recorded in YouTrack

and GitHub. Changes will be reviewed by the other team member before merging.

Expected outcomes

By the end of the diploma project, we expect to deliver:

- A working web application demonstrating the main exchange scenario.
- A documented API and database structure.
- Test results for core functions and access restrictions.
- An evaluation of partner matching against a baseline approach.
- Technical documentation and user instructions.

The evaluation will examine whether the platform helps users find suitable partners. We do not assume in advance that it will improve learning outcomes or that reciprocal matches will always be available.

Current stage

The project is in the analysis and planning stage. An initial README, a GitHub repository, and a YouTrack board have been created.

User interviews, the literature review, comparisons, detailed requirements, architectural diagrams, and the interface prototype are planned work. Application development has not started.

Project links

- GitHub repository
- YouTrack board
- Proposal task SSE-1

2. Relevance evidence and student research

Source document: docs/research/user-interviews.md

Project: Development of a Peer-to-Peer Skill Exchange Platform for University Students

Task: SSE-2 - Interview students about skill exchange

Prepared by: Tamir Burlibaev

Peer reviewer: Nursultan Makhmutov

Repository: Student Skill Exchange

File: `docs/research/user-interviews.md`

Response dates: 10-23 October 2026

Report date: 23 October 2026

Status: Ready for peer review

1. Summary

Six questionnaire responses were analysed to understand how students find learning help and whether a skill-exchange platform could fit their needs. Search by skill was selected by five respondents; offered/wanted skill profiles and the combined proficiency/language filter option were each selected by three.

Five respondents expressed positive interest in using the platform, but only two selected a willingness to receive help and share skills in return. Two wanted to receive help only, and two did not want to participate yet. Interest in the idea therefore cannot be treated as a commitment to reciprocal exchange.

The results support testing skill discovery and clear profiles. They also identify questions about competence, practical arrangements, and teaching readiness that the first prototype should address. This exploratory sample does not establish university-wide demand, successful reciprocal matching, or learning improvement.

2. Method

The study used an online questionnaire in Russian. The analysis includes all six response records, including neutral answers and internally inconsistent selections. The questionnaire replaces the discussion format in the original SSE-2 brief; the repository filename follows the task's required deliverable path.

The source is the six-row response extract, checked against three spreadsheet screenshots. P1-P6 identify the source rows, not chronological order. Dates follow the project owner's correction and supersede the dates displayed in the screenshots. Exact submission times are not used.

Four respondents reported bachelor's study and two reported master's study. Recruitment channel, university affiliation, field of study, response rate, and individual consent selections are not available in the analysis extract. Names and contact details are excluded. The sample should not be presented as representative of Astana IT University or the wider student population.

The notes translate and paraphrase selected answer options. They are not interview quotations. Detailed learning incidents and follow-up explanations were not collected, and the optional comments contained no substantive text.

Measures and counting rules

Response labels D01-D14 are local evidence identifiers. For example, P3-D05 refers to P3's main reported difficulty.

Label	Measure
D01	Self-reported study level
D02	Frequency of needing another person's help with learning in the preceding three months
D03	First method used in the most recent help search during the same three months
D04	Outcome of that search
D05	Main difficulty
D06	Areas in which help is wanted
D07	Possible skills to share
D08	Participation preference
D09	Interaction format
D10	Weekly time budget
D11	Features selected for a first version
D12	Information that would increase comfort with a partner
D13	Hypothetical intention to use the platform
D14	Additional comment

Two distinctions matter for interpreting the history questions. D02 asks how often help was **needed**, not how many searches or sessions took place. D03 asks which method was used **first**, not every channel tried. Both questions refer to the preceding three months relative to the response date.

Counts are the number of respondents selecting an option, with a denominator of six. D06, D07, and D11 permit multiple selections, so their totals can exceed six. No order of preference is inferred within a respondent's selections. The combined proficiency/language option does not measure the importance of those filters separately.

3. Anonymized response records

Participant	Date	Study level
P1	23 October 2026	Bachelor's degree, year 3
P2	19 October 2026	Master's degree
P3	20 October 2026	Bachelor's degree, year 3
P4	15 October 2026	Bachelor's degree, year 2

Participant	Date	Study level
P5	13 October 2026	Bachelor's degree, year 3
P6	10 October 2026	Master's degree

All six records are retained. Answer lists contain reported selections; the separately labelled analyst notes explain limitations or ambiguities. Desired skill categories are not treated as evidence of the subject of a past learning incident.

P1

Date: 23 October 2026

Format: Online questionnaire

- **P1-D01 - Study level:** Bachelor's degree, year 3.
- **P1-D02 - Frequency of needing help:** 3-5 times.
- **P1-D03 - Initial search route:** Asked a student they already knew directly.
- **P1-D04 - Outcome:** Found someone, but could not agree on a session.
- **P1-D05 - Main difficulty:** Did not know where to find a suitable person.
- **P1-D06 - Help wanted:** Programming and software development; Mathematics and statistics; Academic writing and research.
- **P1-D07 - Possible skills to share:** Foreign languages; Video editing; Presentations and public speaking.
- **P1-D08 - Participation preference:** Receive help only.
- **P1-D09 - Interaction format:** Online or in person, depending on the situation.
- **P1-D10 - Weekly time budget:** 1-2 hours per week, inclusive.
- **P1-D11 - Selected features:** Search for students by skill; Filters by proficiency level and communication language; Profile with skills I can teach and want to learn.
- **P1-D12 - Comfort signal:** Reviews from students who have already learned with the person.
- **P1-D13 - Use intention:** Definitely would use.
- **P1-D14 - Additional comment:** No additional comment.

Analyst note: The reason a session could not be arranged was not specified. The offered-skill selections describe possible capabilities; they do not override the receive-help-only participation preference.

P2

Date: 19 October 2026

Format: Online questionnaire

- **P2-D01 - Study level:** Master's degree.
- **P2-D02 - Frequency of needing help:** 6 times or more.
- **P2-D03 - Initial search route:** Looked for a paid tutor or mentor.
- **P2-D04 - Outcome:** Found someone and received suitable help.
- **P2-D05 - Main difficulty:** Found it difficult to assess the person's knowledge.
- **P2-D06 - Help wanted:** Programming and software development; Data analysis and spreadsheets; Presentations and public speaking.
- **P2-D07 - Possible skills to share:** Programming and software development; Mathematics and statistics; Presentations and public speaking.
- **P2-D08 - Participation preference:** Receive help and share skills in return.
- **P2-D09 - Interaction format:** In person only.
- **P2-D10 - Weekly time budget:** More than 2 hours per week.
- **P2-D11 - Selected features:** Search for students by skill; Built-in chat; Choice of a convenient meeting time.
- **P2-D12 - Comfort signal:** Confirmation that the person is a university student.
- **P2-D13 - Use intention:** Probably would use.
- **P2-D14 - Additional comment:** No additional comment.

Analyst note: The paid-help search had a positive reported outcome. Difficulty assessing competence still appeared in this response. Neither the price nor a particular learning incident was described.

P3

Date: 20 October 2026

Format: Online questionnaire

- **P3-D01 - Study level:** Bachelor's degree, year 3.
- **P3-D02 - Frequency of needing help:** 3-5 times.
- **P3-D03 - Initial search route:** Asked a student they already knew directly.
- **P3-D04 - Outcome:** Found someone and received suitable help.
- **P3-D05 - Main difficulty:** Found it difficult to assess the person's knowledge.
- **P3-D06 - Help wanted:** Mathematics and statistics.

- **P3-D07 - Possible skills to share:** Programming and software development; Video editing; Presentations and public speaking.
- **P3-D08 - Participation preference:** Receive help and share skills in return.
- **P3-D09 - Interaction format:** Online or in person, depending on the situation.
- **P3-D10 - Weekly time budget:** Less than 1 hour per week.
- **P3-D11 - Selected features:** Search for students by skill; Filters by proficiency level and communication language; Profile with skills I can teach and want to learn.
- **P3-D12 - Comfort signal:** Reviews from students who have already learned with the person.
- **P3-D13 - Use intention:** Probably would use.
- **P3-D14 - Additional comment:** No additional comment.

Analyst note: Willingness to exchange is accompanied by a limited stated weekly time budget. Actual availability and a suitable teaching level were not collected.

P4

Date: 15 October 2026

Format: Online questionnaire

- **P4-D01 - Study level:** Bachelor's degree, year 2.
- **P4-D02 - Frequency of needing help:** None in the last three months.
- **P4-D03 - Initial search route:** Did not look for help from another person during the period.
- **P4-D04 - Outcome:** Did not look for help from another person.
- **P4-D05 - Main difficulty:** Did not look for help from another person.
- **P4-D06 - Help wanted:** Unsure.
- **P4-D07 - Possible skills to share:** Unsure.
- **P4-D08 - Participation preference:** Do not want to participate yet.
- **P4-D09 - Interaction format:** Online only.
- **P4-D10 - Weekly time budget:** Cannot estimate yet.
- **P4-D11 - Selected features:** Cannot choose yet.
- **P4-D12 - Comfort signal:** Unsure.
- **P4-D13 - Use intention:** Might or might not use.
- **P4-D14 - Additional comment:** No additional comment.

Analyst note: This response does not demonstrate a recent help-seeking problem or current willingness to participate. The online preference is a stated preference, not evidence of actual use.

P5

Date: 13 October 2026

Format: Online questionnaire

- **P5-D01 - Study level:** Bachelor's degree, year 3.
- **P5-D02 - Frequency of needing help:** None in the last three months.
- **P5-D03 - Initial search route:** Asked a student they already knew directly.
- **P5-D04 - Outcome:** Found someone and received suitable help.
- **P5-D05 - Main difficulty:** Did not know where to find a suitable person.
- **P5-D06 - Help wanted:** Data analysis and spreadsheets; Graphic design; Another field, not specified.
- **P5-D07 - Possible skills to share:** Foreign languages; Graphic design; Not ready to explain skills to others yet.
- **P5-D08 - Participation preference:** Do not want to participate yet.
- **P5-D09 - Interaction format:** In person only.
- **P5-D10 - Weekly time budget:** Do not plan to allocate time.
- **P5-D11 - Selected features:** Search for students by skill; Profile with skills I can teach and want to learn; Send, accept, and decline exchange requests.
- **P5-D12 - Comfort signal:** Examples of the person's work or projects.
- **P5-D13 - Use intention:** Definitely would use.
- **P5-D14 - Additional comment:** No additional comment.

Analyst note: P5 reported no occasions of needing help but also reported a help search and suitable help in the same three-month recall period. This inconsistency is unresolved. Both selections are retained as reported, without treating the search or its outcome as confirmed recent history. Possible sharing areas were selected together with not being ready to teach. Definite hypothetical interest coexists with no current participation or planned time.

P6

Date: 10 October 2026

Format: Online questionnaire

- **P6-D01 - Study level:** Master's degree.
- **P6-D02 - Frequency of needing help:** Do not remember.

- **P6-D03 - Initial search route:** Used another method, not specified.
- **P6-D04 - Outcome:** Received help, but it did not resolve the question.
- **P6-D05 - Main difficulty:** Felt uncomfortable asking for help.
- **P6-D06 - Help wanted:** Graphic design; Academic writing and research; Presentations and public speaking.
- **P6-D07 - Possible skills to share:** Programming and software development; Mathematics and statistics; Data analysis and spreadsheets.
- **P6-D08 - Participation preference:** Receive help only.
- **P6-D09 - Interaction format:** Online or in person, depending on the situation.
- **P6-D10 - Weekly time budget:** More than 2 hours per week.
- **P6-D11 - Selected features:** Search for students by skill; Filters by proficiency level and communication language; Notifications about requests and meetings.
- **P6-D12 - Comfort signal:** Recommendation from someone they know.
- **P6-D13 - Use intention:** Probably would use.
- **P6-D14 - Additional comment:** No additional comment.

Analyst note: The alternative search method and the learning problem were not specified. Possible teaching areas do not establish willingness to teach when the participation preference is receive-help-only.

4. Descriptive results

Learning help and search outcomes

Measure	Selections and participants
Frequency of needing help	3-5 times: 2 (P1, P3); 6 or more: 1 (P2); none: 2 (P4, P5); do not remember: 1 (P6).
First route in the most recent search	A known student: 3 (P1, P3, P5); paid tutor or mentor: 1 (P2); another unspecified method: 1 (P6); no search: 1 (P4).
Outcome	Suitable help: 3 (P2, P3, P5); found someone but no session agreed: 1 (P1); help did not resolve the question: 1 (P6); no search: 1 (P4).
Main difficulty	Knowing where to find a suitable person: 2 (P1, P5); assessing knowledge: 2 (P2, P3); discomfort asking: 1 (P6); no search: 1 (P4).

These are counts of selected answers. P5 reported no need for help but also reported a search and suitable help in the same recall period. The history selections remain in the tables, with this inconsistency unresolved; they are not treated as confirmed recent events. P5's other preferences are retained rather than discarding the entire response.

Participation and practical preferences

Measure	Selections and participants
Participation	Receive help and share skills in return: 2 (P2, P3); receive help only: 2 (P1, P6); do not want to participate yet: 2 (P4, P5).
Interaction format	Online or in person depending on the situation: 3 (P1, P3, P6); in person only: 2 (P2, P5); online only: 1 (P4).
Weekly time budget	More than 2 hours: 2 (P2, P6); 1-2 hours inclusive: 1 (P1); less than 1 hour: 1 (P3); cannot estimate: 1 (P4); no planned time: 1 (P5).
Use intention	Definitely would use: 2 (P1, P5); probably would use: 3 (P2, P3, P6); might or might not use: 1 (P4).

Among the five positive use-intention responses, P2 and P3 selected reciprocal participation, P1 and P6 wanted help only, and P5 did not currently want to participate or allocate time. These measures must remain separate.

Features selected for a first version

Selected option	Count out of 6	Participants
Search for students by skill	5	P1, P2, P3, P5, P6
Filters by proficiency level and communication language	3	P1, P3, P6
Profile with skills I can teach and want to learn	3	P1, P3, P5
Built-in chat	1	P2
Cannot choose yet	1	P4
Choice of a convenient meeting time	1	P2
Notifications about requests and meetings	1	P6
Send, accept, and decline exchange requests	1	P5

Search had the broadest support in this sample. The other feature counts are preferences to investigate, not a complete specification. A feature selected once may still be necessary for the agreed workflow; a feature selected by several people still needs usability and scope review.

Information that would increase comfort

Selected option	Count out of 6	Participants
Reviews from students who have already learned with the person	2	P1, P3
Confirmation that the person is a university student	1	P2
Examples of the person's work or projects	1	P5
Recommendation from someone they know	1	P6
Unsure	1	P4

No single comfort signal was selected by a majority. Student-status confirmation and evidence of subject competence address different questions. Selecting reviews does not establish a need for numerical public ratings.

Learning interests and possible contributions

Area	Help wanted	Possible skills to share
Programming and software development	2 (P1, P2)	3 (P2, P3, P6)
Mathematics and statistics	2 (P1, P3)	2 (P2, P6)
Data analysis and spreadsheets	2 (P2, P5)	1 (P6)
Academic writing and research	2 (P1, P6)	0
Presentations and public speaking	2 (P2, P6)	3 (P1, P2, P3)
Graphic design	2 (P5, P6)	1 (P5)
Foreign languages	0	2 (P1, P5)
Video editing	0	2 (P1, P3)
Another field, not specified	1 (P5)	0

Values are counts with participant IDs. Zero means that no respondent selected the area in that column. P4 was unsure about both learning interests and possible contributions. P5 selected possible skills alongside not being ready to explain them to others. P1 and P6 also listed possible skills while choosing to receive help only. These categories are not verified qualifications or commitments to teach.

5. Evidence-based findings

F01 - Skill-based search was the most selected feature. Five selected search by skill (P1/P2/P3/P5/P6-D11). P1 and P5 selected difficulty knowing where to find a suitable person (D05), although P5's history is inconsistent. This supports testing discovery, not a claim that all students have the same problem.

F02 - Competence and comfort need attention. P2 and P3 found it difficult to assess knowledge (D05), despite reporting suitable help. Comfort signals varied: reviews for P1 and P3, student status for P2, work samples for P5, and a personal recommendation for P6 (D12). No trust mechanism was tested.

F03 - Finding someone did not always resolve the learning need. P1 found someone but could not agree on a session; P6 received help that did not resolve the question (D04). The responses do not establish whether timing, expectations, teaching ability, or another factor caused these outcomes.

F04 - Interest and reciprocal participation are different. Five expressed positive use intention (D13), but only P2 and P3 chose to both receive and provide help (D08). P5's definite interest coexists with no current participation or planned time. Skills listed in D07 must not be interpreted as teaching commitments.

F05 - Format and time preferences vary. The responses include flexible, in-person-only, and online-only preferences, together with several time-budget categories (P1-P6-D09/D10). These are stated preferences, including those of respondents not ready to participate; actual shared availability remains unknown.

F06 - Profiles and filters are candidates for testing. Offered/wanted profiles were selected by P1, P3, and P5; the combined level/language filter option by P1, P3, and P6 (D11). This supports evaluating those designs. It does not establish that each filter is essential or that the profile produces a suitable reciprocal match.

6. Team assumptions

The following interpretations are kept separate from the reported answers.

Assumption	Assessment	Evidence or remaining question
A01 - Some students struggle to find suitable learning help	Partial support in this sample.	P1 and P5 selected difficulty finding where to look; P5's history needs clarification. Prevalence is unknown.
A02 - Most interested students will teach in return	Not supported.	Only P2 and P3 selected reciprocal participation. Positive use intention is broader than willingness to teach.
A03 - Broad skill categories are enough for useful reciprocal matching	Not established.	No topic-level proficiency, accepted match, or completed exchange was observed.
A04 - All proposed filters are essential	Not established.	The combined level/language option was selected three times; individual filter value and mandatory filtering were not tested.
A05 - Public ratings are required in the MVP	Not established.	Two selected reviews. A rating system, its alternatives, and its moderation needs were not evaluated.
A06 - Use intention predicts sustained use and learning benefit	Not established.	The survey contains hypothetical intentions, not observed use, repeat participation, or learning outcomes.

7. Implications for the platform

The proposed decisions below connect the response evidence with the project direction. R01-R08 are local labels, not new YouTrack issues. Implementation priorities must be reconciled with SSE-6.

Proposed decision	Evidence	Action and boundary
R01 - Prioritise skill-based search	F01; P1/P2/P3/P5/P6-D11	Test whether students can find relevant help by a concrete topic. Measure relevance and ease of discovery, not just search-result volume.
R02 - Keep offered and wanted skills separate	F04, F06; P1/P3/P5-D11; D06/D07	Use clear skill descriptions and explicit teaching readiness. Do not automatically publish every listed capability as an active teaching offer.
R03 - Test transparent direct reciprocal suggestions	F04; A03; SSE-3	Retain simple two-way matching and short explanations as the project approach. The survey does not validate the algorithm. Confirm topic compatibility and mutual interest in a pilot.
R04 - Make requests and agreement explicit	F03, F04; P1-D04; P5-D11; SSE-3	Retain request, accept, and decline actions. State what each person offers and wants before acceptance. Check cancellation, completion, and rescheduling rules against SSE-6.
R05 - Validate filters and session arrangements	F05, F06; P1/P3/P6-D11; P2-D11	Test language and level separately. Support clear agreement on time and format within the agreed scope. Availability matching and calendar integration require further justification.
R06 - Test useful partner information before public ratings	F02; P2/P3-D05; D12	Evaluate concise experience descriptions and work samples. Keep student status distinct from competence. Treat reviews and reputation as future candidates under SSE-3.
R07 - Separate browsing from agreement to teach	F04; P1/P6-D07/D08; P5-D07/D08/D10/D13	Make participation voluntary and clarify what happens when no reciprocal match exists. Whether one-way help is supported remains a product-scope decision.

Proposed decision	Evidence	Action and boundary
R08 - Review extra communication features against scope	P2-D11; P6-D11	Chat and notifications were each selected once. Compare them with simpler arrangements and the SSE-6 scope rather than adding them automatically.

Consistency with the literature review

The team's SSE-3 literature review, sections 6 and 7, provides the design baseline: offered/wanted profiles, search, direct reciprocal matches with brief explanations, and voluntary exchange requests. This survey adds local preference evidence and exposes the gap between interest and willingness to reciprocate.

Availability, language, proficiency, and format filters still need focused validation. Public ratings and machine-learning or graph-based ranking remain future candidates; formal grading remains outside the current scope. The questionnaire did not evaluate those mechanisms. These scope decisions follow SSE-3 and are distinct from the survey findings.

8. Limitations and next validation

The study is exploratory. Six responses with an undocumented recruitment process cannot establish population-level demand. The answers are self-reports based largely on closed options, with no detailed incident narratives or follow-up explanations. Broad categories do not establish proficiency, teaching readiness, or a viable match.

P5's history remains inconsistent and is preserved with that qualification. The combined filter option does not separate language from proficiency. Hypothetical use intention and stated time budgets are not evidence of actual participation. Recruitment details, university affiliations, consent selections, and the form URL were not included in the analysis record.

The next checks should be practical:

1. Clarify P5's history and teaching readiness if follow-up is possible, keeping the original response and clarification as separate records.
2. Collect concrete learning examples to understand P1's failed agreement and P6's unresolved help, rather than assuming their causes.
3. Test skill search and profiles with realistic topics. Check whether two willing students can identify a useful reciprocal match and agree on time and format.
4. Record accepted and completed exchanges, usefulness to each participant, unmatched cases, and willingness to return.

9. Conclusion

The responses support a focused prototype for finding learning help and understanding what a potential partner can offer. They also show that reciprocal exchange needs direct validation: five expressed positive intention to use the platform, while only two selected willingness to help in return. The project should test the complete exchange journey before claiming a working reciprocal community or a learning benefit.

10. Review and integration

The research content is ready for peer review. The original SSE-2 brief describes 3-5 discussions; its description and acceptance checklist should reflect the questionnaire method and actual six-response sample.

- [ ] Nursultan has reviewed this revision through a pull request.
- [ ] Review comments have been addressed and the revision merged into `main`.
- [ ] The pull request is linked to SSE-2.

Sources

1. Student questionnaire response extract: six anonymized records, checked against the three spreadsheet screenshots. Dates corrected by the project owner to 23, 19, 20, 15, 13, and 10 October 2026 in source-row order.
2. Team document, SSE-3: *Literature Review: Peer-to-Peer Skill Exchange for University Students*, especially sections 6 and 7, supplied on 23 October 2026.

3. Literature review

Source document: docs/research/literature-review.md

Project: Development of a Peer-to-Peer Skill Exchange Platform for University Students

Task: SSE-3

Status: Ready for team review

Review type: Targeted narrative review with a structured source analysis

1. Purpose and review scope

The proposed platform will help university students describe the skills they can teach, specify the skills they want to learn, discover potentially compatible peers, and request a learning exchange. This review examines three bodies of research that are directly relevant to that goal:

1. peer learning in higher education;
2. reciprocal, people-to-people recommendation;
3. motivation, trust, reputation, and interface transparency.

The review combines foundational publications with recent work published from 2021 to 2023. Bibliographic details were checked against DOI records or official publisher and proceedings pages. The review is targeted rather than systematic: it does not claim to identify every study in these fields. Its purpose is to provide an evidence-informed basis for the first product requirements and to identify assumptions that still require validation with target users.

2. Peer learning in higher education

Peer learning covers several different arrangements rather than one uniform teaching method. Topping (1996) distinguishes models by factors such as participants' year of study, group size, role continuity, place, time, and learning objective. His review supports the value of several structured forms of peer tutoring, but it does not imply that every informal student-to-student interaction will improve learning. The design and context of the activity remain important.

The same distinction appears in the study by Topping et al. (1996). Forty-five first-year calculus students completed twelve one-hour paired-tutoring sessions. Their examination performance was better than that of the previous year's cohort, and students reported gains in transferable skills. However, the cohorts were not equivalent, so the study cannot establish that peer tutoring alone caused the difference. The study is useful for showing what a structured exchange may look like, while its design cautions against making strong causal claims.

Boud, Cohen, and Sampson (1999) focus on the relationship between peer learning and assessment. They argue that assessment can support collaboration through peer feedback and shared responsibility, but it can also undermine collaboration when only individual achievement is rewarded. This is relevant to the platform because feedback should support reflection and coordination rather than turn every exchange into formal grading.

Recent evidence strengthens the case for studying peer learning while preserving this caution. Bengesai, Amusa, and Dhunpath (2023) screened 1,645 records and included 37 controlled or comparison-group studies in a meta-analysis of formal peer-learning approaches in undergraduate education. They found a moderately strong positive relationship with course performance, while also noting geographical concentration, substantial representation of STEM subjects, and a need for more context-specific experimental evidence.

Two recent studies address online environments. Tang, Lau, and Chau (2022) surveyed 204 sub-degree students who had participated in an online peer-learning revision centre. Factor and correlation analyses connected learning ability, attitude, motivation, and interpersonal relationships, but the self-report design and single setting limit generalisation. Mendieta-Aragón et al. (2023) studied 78 engineering students who created or consumed peer-produced videos in a hybrid university course. Participants evaluated the activity positively, especially those who acted as peer teachers. Participation was voluntary, and outcomes relied substantially on student perceptions, so the findings show acceptance and perceived value rather than a definitive causal effect on achievement.

Taken together, these studies support a platform that makes learning goals, offered skills, requested skills, and participant roles explicit. They also support voluntary participation and simple guidance for arranging an exchange. They do not establish that a matching platform will automatically improve academic performance; that outcome must be evaluated after the platform is used.

3. Reciprocal matching and recommendation

Traditional recommender systems suggest passive items to a user. A people-to-people system is different because a useful match depends on both participants. Pizzato et al. (2010) formalised this distinction and identified applications such as finding a mentor, business partner, employee, or date. Their work is conceptual and illustrated through online dating, but the core requirement transfers to skill exchange: the proposed relationship should be acceptable and useful to both students.

Vitale, Parotsidis, and Gentile (2018) formalised online reciprocal recommendation as a sequential-learning problem. Their algorithms were evaluated on synthetic and real-world datasets and performed better than simple baselines under stated assumptions. The work shows why mutual interest must be modelled explicitly, but it does not provide a directly deployable algorithm for a new university platform with no interaction history.

Neve and Palomares (2019) studied how two directional preference scores can be combined into one compatibility score. Their experiments show that the aggregation rule affects precision and F1 performance; in their setting, a cross-ratio approach outperformed the harmonic mean. This result is useful as evidence that a reciprocal score should not be chosen arbitrarily, although it comes from one model and dataset rather than a university skill-exchange deployment.

Palomares et al. (2021) review the broader reciprocal-recommender literature. They identify preference prediction, fusion of two one-sided preferences, evaluation, data sparsity, and application-specific requirements as central issues. Chang et al. (2022) provide a more recent graph-neural-network approach for reciprocal friend recommendation in an online gaming service. Their model separates preference, attractiveness, and similarity and performs well on large real-world datasets. Its data and infrastructure needs are far beyond those of an initial university prototype.

The appropriate first implementation is therefore a transparent rule-based match. A direct reciprocal match exists when Student A offers a skill that Student B wants and Student B offers a skill that Student A wants. Additional filters, such as availability or preferred session format, should be included only if student research shows that they materially affect partner choice. Machine-learning and graph-based ranking should remain future work until the platform has sufficient interaction data and a validated evaluation metric.

4. Motivation, trust, reputation, and transparency

Self-determination theory identifies autonomy, competence, and relatedness as basic psychological needs associated with self-directed motivation and well-being (Ryan & Deci, 2000). Applied cautiously, this suggests that students should control their learning goals, decide which exchanges to accept, present skills they feel competent to teach, and interact with peers rather than being assigned automatically. The theory provides a design rationale, not proof that these interface choices will cause sustained use.

Resnick et al. (2000) explain how reputation systems collect and distribute information about past behaviour to support interaction between people who do not already know each other. The article provides a foundation for considering reviews or participation history, but it also concerns internet transactions rather than university learning. A rating feature may introduce cold-start, bias, retaliation, and moderation problems. It should not be treated as a required MVP feature without user and supervisor approval.

Kizilcec (2016) tested three levels of interface transparency in an online field experiment involving peer assessment. When a result violated participants' expectations, an explanation could restore trust, but excessive information reduced trust. The practical lesson is balance: the platform should provide a short, concrete reason for a recommendation, such as “You want to learn Python; this student offers Python and wants to learn English, which you offer.” It should not expose users to an opaque score or a long technical explanation.

5. Structured source analysis

Source	Method and context	Main finding used in this project	Important limitation	Project implication
Topping (1996)	Typology and review of peer tutoring in further and higher education	Structured peer-tutoring models can support learning	Broad and dated review; outcomes vary by design and context	Define skills, goals, roles, and exchange format clearly
Boud et al. (1999)	Conceptual analysis of peer learning and assessment	Assessment design can support or undermine collaboration	Does not evaluate a digital matching platform	Keep feedback developmental; avoid compulsory grading in the MVP
Topping et al. (1996)	Paired-tutoring study with 45 calculus students and 12 sessions	Structured paired tutoring was associated with better results and reported transferable skills	Previous-year comparison group was non-equivalent	Offer a simple structure for planning sessions; evaluate outcomes cautiously
Ryan & Deci (2000)	Theoretical review of self-determination research	Autonomy, competence, and relatedness are central to self-directed motivation	General psychological theory, not a platform trial	Let students choose goals, skills, and partners and accept or reject requests
Resnick et al. (2000)	Conceptual article on online reputation systems	Visible information about prior behaviour can support trust among strangers	Commercial transaction focus; reputation can be biased or manipulated	Treat reviews/reputation as a validated future feature, not an automatic MVP requirement
Pizzato et al. (2010)	Conceptual workshop paper with an online-dating case study	People-to-people recommendation must consider both parties	Early work; domain differs from education	Match offered and wanted skills in both directions
Kizilcec (2016)	Online field experiment with three transparency levels in peer assessment	A concise explanation can repair trust; too much information can reduce it	Peer-assessment context and expectation-violation condition	Show a brief human-readable reason for each suggested match
Vitale et al. (2018)	Theoretical sequential-learning model tested on synthetic and real data	Reciprocal matching differs formally from one-sided recommendation	Relies on structural assumptions and existing interaction data	Do not use a one-sided recommendation score; postpone learning-based ranking
Neve & Palomares (2019)	Comparative experiment on bidirectional-score aggregation	The choice of aggregation function changes recommendation quality	One model and dataset; no education-specific validation	If ranking is added later, compare aggregation methods rather than selecting one without evidence
Palomares et al. (2021)	State-of-the-art review of reciprocal recommender systems	Mutual compatibility requires preference prediction, fusion, and suitable evaluation	Survey spans domains with different goals and risks	Define success as an accepted, useful match for both students
Tang et al. (2022)	Survey of 204 participants in an online peer-learning revision centre	Learning, motivation, attitudes, and interpersonal factors were closely related	Self-report, single context, no non-participant control group	Measure motivation and relationship quality during evaluation, not only number of matches
Chang et al. (2022)	Multiplex-graph neural network evaluated on online-gaming data	Preference, attractiveness, and similarity can be modelled separately at scale	Complex, data-intensive, and outside the education domain	Keep GNN-based matching outside the MVP; revisit only after sufficient data exists
Mendieta-Aragón et al. (2023)	Survey-based study of 78 engineering students in hybrid learning	Students valued digital peer learning; peer teachers reported motivational and skill benefits	Voluntary sample and subjective outcomes limit causal inference	Make exchanges voluntary and evaluate teacher and learner experiences separately

Source	Method and context	Main finding used in this project	Important limitation	Project implication
Bengesai et al. (2023)	Meta-analysis of 37 undergraduate studies selected from 1,645 records	Formal peer learning has a moderately strong positive relationship with course performance	Evidence is concentrated geographically and in STEM; interventions vary	Peer learning is plausible, but effectiveness must be tested in the local university context

6. Research gap

The literature supports structured peer learning and shows that people-to-people matching should account for both participants. It also offers design principles for autonomy, trust, and explanation. However, none of the reviewed studies establishes that students at Astana IT University currently have a substantial problem finding skill-exchange partners or that they would repeatedly use a dedicated platform.

The main project gap is therefore contextual and integrative. Existing work studies peer-learning interventions, reciprocal algorithms, or trust mechanisms separately and usually in other institutions or domains. The proposed project will test whether a simple combination of these ideas is useful for the target students.

SSE-2 should validate the following assumptions:

- students currently find it difficult or time-consuming to locate suitable partners;
- students are willing both to teach and to learn;
- offered and wanted skills are sufficient for initial partner discovery;
- availability, language, session format, or proficiency level materially influence partner choice;
- students are comfortable contacting peers through the proposed request flow;
- students want post-exchange feedback and understand how it would be used.

7. Evidence-informed scope decisions

Decision	Status	Rationale
Profiles with offered and wanted skills	Include in MVP	Required for structured peer learning and reciprocal matching
Search and direct reciprocal matches	Include in MVP	Implements mutual compatibility without requiring historical data
Short explanation for a suggested match	Include in MVP	Supports appropriate transparency without information overload
Request, accept, or decline an exchange	Include in MVP	Preserves autonomy and confirms mutual interest
Availability, language, proficiency, and session-format filters	Validate through SSE-2	Plausible constraints, but their importance to target users is not yet established
Ratings or public reputation scores	Future candidate	May support trust but introduces cold-start, bias, moderation, and retaliation risks
Machine-learning or graph-based ranking	Future candidate	Requires interaction data, evaluation metrics, and more implementation complexity
Formal peer assessment or grading	Outside current scope	The project supports skill exchange rather than academic assessment

8. Conclusion

The evidence supports three design principles. First, peer learning works best when goals, roles, and activities are structured. Second, partner discovery is a reciprocal problem: a useful match must account for both students. Third, autonomy and concise explanations should be built into the interaction, while ratings and complex recommendation algorithms should be added only after validation.

These findings justify a small, transparent MVP centred on profiles, offered and wanted skills, reciprocal partner discovery, and exchange requests. They do not prove local demand or educational effectiveness. Those claims must be tested through the SSE-2 questionnaire and later usability and outcome evaluation.

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4. Existing systems comparison

Source document: docs/research/existing-systems-comparison.md

Project: Development of a Peer-to-Peer Skill Exchange Platform for University Students

YouTrack task: SSE-4

Project board: Student Skill Exchange Kanban

Required repository path: `docs/research/existing-systems-comparison.md`

Document owner / task assignee: Tamir Burlibaev

Required peer reviewer: Nursultan Makhmutov

Research and access date: 23 October 2026

Confirmed task deadline: 23 October 2026, 20:00, Astana time (UTC+5)

Repository: HashiraOfSun/student-skill-exchange

Status: Research document prepared; peer review, merge into `main`, and linking the PR to YouTrack remain pending.

1. Purpose and method

This comparison examines four existing platforms using the same ten criteria. It supports decisions about profiles, finding a learning partner, arranging an exchange, and the scope of our first version.

The evidence comes from official product pages, help articles, and public service listings retrieved on 23 October 2026. It is a review of documented capabilities. No accounts were created, sessions booked, payments made, or authenticated workflows tested. Provider statements establish what a platform describes, rather than the quality or reliability of its operation.

The following labels apply throughout the comparison:

- **Documented:** the cited material explicitly describes or displays the capability.
- **Unknown:** the reviewed material does not establish the capability or its details. This does not mean the feature is absent.
- **Team interpretation / proposal:** our reasoning or a candidate design decision, kept separate from product facts.

Sources are linked through the IDs in each table cell. Section 7 records a URL and access date for every source. Several Simbi pages were inconsistently retrievable; the relevant source notes explain the evidence available. No numerical ranking is assigned because the platforms serve different purposes and were not tested under comparable conditions.

2. Why these four systems were selected

Platform	Documented focus	Reason for selection
Simbi	A general network for exchanging services, with direct exchanges and internal credits. [S1], [S2]	The closest example of a broad exchange model. It helps us consider what happens when two people do not need each other's services at the same time.
Tandem	Reciprocal language practice through conversations with other learners and speakers. [T1]	A focused example of expressing what someone can offer and wants to learn, then finding a compatible partner.
Knack	University-based, course-specific peer tutoring, supported through institutional programs. [K1]	Directly relevant to student communities and to arranging academic help around university courses.
ADPList	Sessions with experienced mentors in areas including design, engineering, product, and career development. [A1], [A2]	Relevant to students seeking practical feedback. It provides a useful comparison between reciprocal peer exchange and learning from an experienced mentor.

Together, these systems cover broad service exchange, language exchange, campus tutoring, and professional mentorship. This is a purposive sample, rather than an exhaustive survey of the market.

3. Consistent comparison criteria

ID	Criterion	Question used for every platform
C1	User profiles	What information helps a learner understand the person offering help?
C2	Offered and desired skills	How are what a person can teach and what they want to learn represented?
C3	Search and discovery	How can a user find relevant people, offers, or requests?
C4	Partner matching	What connects learning needs with someone able to help?
C5	Request and agreement workflow	How is a proposed interaction initiated and agreed?
C6	Scheduling	How are availability, time, duration, and meeting format arranged?
C7	Communication and learning format	How can participants communicate or conduct a session?
C8	Pricing and exchange model	Who pays, or what is exchanged?
C9	Trust and safety	What documented information or controls support an informed and safer choice?
C10	University community support	Is there documented support for students or institutions, and what remains unverified?

4. Comparison matrix

The matrix is divided into two tables for readability. Both use the same four platforms and the criteria defined above.

4.1 Profiles, discovery, and agreement

Criterion	Simbi	Tandem	Knack	ADPList
C1. User profiles	**Documented:** public provider identity, qualifications, availability notes, and reviews appear on a sampled service page. [S3]	**Documented:** photos, languages spoken and being learned, interests, and learning preferences. [T2]	**Documented:** tutor photo, biography, background, relevant experience, and availability. [K2]	**Documented:** mentor profiles show professional roles, expertise, and reviews; mentors can add a biography. [A1], [A3]

Criterion	Simbi	Tandem	Knack	ADPList
C2. Offered and desired skills	**Documented:** service offers and separate requests. A paired, structured teach/learn profile is **Unknown**. [S2], [S4]	**Documented:** spoken and target languages represent the two sides of an exchange. General subject-to-subject exchange is **Unknown**. [T1], [T2]	**Documented:** active tutoring courses and course-help requests. Reciprocal skill-pair declarations are **Unknown**. [K3]	**Documented:** mentors specify topics they can help with; mentors may also book learning sessions. Paired reciprocal skill fields are **Unknown**. [A3]
C3. Search and discovery	**Documented:** categories and public service listings. Specific text-search and filter controls are **Unknown** from the retrieved pages. [S2]	**Documented:** discovery by language, location, and interests; some advanced discovery options belong to Pro. [T1], [T2]	**Documented:** tutors see requests relevant to their active courses. [K3]	**Documented:** browse by expertise, inspect profiles and reviews, and use search filters. [A2]
C4. Partner matching	**Documented:** members can browse and propose exchanges; credits enable indirect exchange. Automated reciprocal ranking is **Unknown**. [S1], [S2]	**Documented:** the community feed considers language combinations. The exact ranking method is **Unknown**. [T2]	**Documented:** learners request course help and interested tutors respond. This is course-based connection, with no reciprocal skill algorithm confirmed. [K1], [K3]	**Documented:** goal-aligned mentor suggestions and user choice. The ranking method and reciprocal skill matching are **Unknown**. [A1]
C5. Request and agreement workflow	**Documented:** conversations and exchange proposals. Detailed acceptance, cancellation, and completion states are **Unknown**. [S1]	**Documented:** users initiate conversations. A separate structured exchange-request lifecycle is **Unknown**. [T2]	**Documented:** session requests can be accepted or changed; eligible requests may use Instant Booking. [K4]	**Documented:** choose a mentor and book a time slot. Approval and cancellation rules for every session type are **Unknown**. [A2]

4.2 Arranging a session, cost, trust, and campus context

Criterion	Simbi	Tandem	Knack	ADPList
C6. Scheduling	**Documented:** a sampled listing includes availability notes and asks users to arrange timing in advance. Built-in calendar booking is **Unknown**. [S3]	**Unknown:** a dedicated availability calendar or booking workflow was not established by the reviewed feature pages. [T1], [T2]	**Documented:** date, time, expected duration, location, availability, and once/weekly scheduling are part of session requests. [K4]	**Documented:** mentors publish availability and session length; learners select an available calendar slot. [A2], [A3]
C7. Communication and learning format	**Documented:** the sampled listing mentions communication on Simbi and optional external channels. Built-in video is **Unknown**. [S3]	**Documented:** text, voice notes, audio/video calls, corrections, and translation tools. [T1]	**Documented:** messaging, in-person sessions, and a built-in Online Classroom; group sessions are also described. [K1], [K4]	**Documented:** live one-to-one video sessions and post-session notes. [A1], [A3]
C8. Pricing and exchange model	**Documented:** free joining and cashless exchanges; direct trades or internal credits, which have no dollar value. [S1], [S2], [S4]	**Documented:** a free version and a paid monthly/yearly Pro subscription. Exact regional and university-partnership prices are **Unknown**. [T2], [T4]	**Documented:** institutions fund tutoring for eligible learners. Coverage can be limited to certain students or courses. Institutional fees are **Unknown**. [K1]	**Documented:** free sessions are available and some mentors offer paid sessions. Exact booking fees are **Unknown**; official pages contain inconsistent free-booking wording. [A3], [A4]
C9. Trust and safety	**Documented:** public reviews and self-described qualifications. Credential verification, reporting, and moderation controls are **Unknown** in the inspected pages. [S3], [S4]	**Documented:** application review, authentic-profile rules, reporting, moderation, and an appeals process. Their effectiveness was not tested. [T1], [T3]	**Documented:** course approval and possible training requirements govern tutor activation. Additional identity and moderation controls are **Unknown** here. [K3]	**Documented:** mentor applications are manually reviewed and previous mentees can leave reviews. Detailed identity-check methods are **Unknown**. [A2], [A3]
C10. University community support	**Documented:** a general community network. University verification, campus-only membership, and institutional administration are **Unknown**. [S2], [S4]	**Documented:** university partnerships, campus rollouts, onboarding support, and optional usage reporting. Campus-only membership and university sign-in are **Unknown**. [T4]	**Documented:** school-specific courses and peer tutors, with institutional eligibility rules. Availability at Astana IT University is **Unknown**. [K1], [K4]	**Documented:** students are included among the intended mentees. Current university-only networks and institutional sign-in are **Unknown** from these sources. [A2]

4.3 Important qualifications

ADPList pricing: the mentor recruitment page contains a general statement that booking a mentor is free. Its comparison guide, updated on 15 October 2026, explicitly describes both free and paid sessions. We retain that distinction and leave exact charges unverified; the price and terms of a particular booking need to be checked before use. [A3], [A4]

Knack access and booking: free access depends on institutional coverage, rather than being guaranteed to every university student. Its documentation also distinguishes course-help requests from scheduling a particular session. Instant Booking describes confirmation of eligible session requests; it should not be read as a guarantee of immediate tutoring. [K1], [K3], [K4]

University support: Tandem explicitly advertises an institutional offering, so it would be inaccurate to describe all non-tutoring platforms as lacking university support. Campus-only access or identity verification requires separate evidence. [T4]

Simbi retrieval: the service catalog, organizational profile, and a sampled service page supplied usable public evidence. The How It Works description was available as an official indexed extract, while direct retrieval was unsuccessful. Some other Simbi requests also returned access errors. These observations limit verification; they do not establish missing features or a platform-wide outage. [S1], [S2], [S3], [S4]

5. What existing systems address

The observations below are product facts followed by our interpretation of their relevance.

- **Exchanging help without an immediate reciprocal match:** Simbi documents internal credits as well as direct exchange. **Team interpretation:** an exchange design needs an explicit policy for cases where one person's offer does not interest the other. Credits are one possible approach, with additional accounting and dispute-handling decisions. [S1], [S2]
- **Finding someone with complementary knowledge:** Tandem makes spoken and target languages visible and uses language combinations in discovery. **Team interpretation:** showing both sides of a learning relationship can guide our profile and matching design. Whether the same approach works across unrelated subjects needs testing. [T1], [T2]
- **Organising academic support inside a university:** Knack connects university courses, peer tutors, and session arrangements. **Team interpretation:** campus context and concrete session details deserve separate attention; the institutional funding model is a different commitment from a student-run prototype. [K1], [K4]
- **Getting feedback from a more experienced person:** ADPList supports choosing a professional mentor and booking a session. **Team interpretation:** mentor-led help is a relevant alternative when a student wants feedback rather than a reciprocal exchange. Our service must offer a useful reason to choose peer exchange in that situation. [A1], [A2]

The comparison documents several workable approaches. It does not establish how frequently our intended users face a problem, how satisfied they are with existing options, or whether they will adopt another platform. Those questions belong in real student interviews and prototype evaluation.

6. Proposed contribution and requirement implications

Our repository describes a first version with student profiles, offered and desired skills, skill search, suggested partners, exchange requests, and agreement on a topic, time, and format. Built-in video, payments, a native mobile app, and machine learning are outside that stated scope. The README was observed at the repository state associated with commit `6234b14e69aa693151e0fe9e61dcbb54e2bdd77e`. [P1]

The team's SSE-3 literature review supports a small first version with transparent reciprocal matching and voluntary exchanges. The decisions below connect that direction with the product comparison. They are **team proposals**, to reconcile with SSE-6 requirements and evidence from SSE-2 student research. The value of the proposed combination remains a hypothesis to evaluate, rather than a claim of market-wide novelty.

6.1 Profiles and reciprocal discovery in the MVP

Keep offered and wanted skills separate. Provide skill search and simple rules that check both directions: Student A offers what Student B wants, and Student B offers what Student A wants. Show a short explanation, such as a Python-for-English exchange, and let students choose. Tandem's two language roles provide a useful product reference. [T2]

6.2 Exchange requests in the MVP

Support a request with the proposed help in both directions, followed by acceptance or rejection. Record the agreed topic, time, and format. Knack's concrete session requests are a useful reference. Check cancellation, completion, and rescheduling rules against SSE-6 before implementation. [K4]

6.3 Filters that need student validation

Knack and ADPList show availability and session details, but that does not establish which discovery filters our students need. Validate availability, language, proficiency, and session-format filters through SSE-2. Recording an agreed time and format does not require an availability calendar or calendar integration. [K4], [A3]

6.4 Trust measures and future ratings

Profiles, reviews, and approval rules provide different kinds of information about a partner. For the pilot, propose clear experience descriptions, deliberate contact sharing, and a basic reporting procedure with an assigned owner. Treat self-reported ability separately from verified credentials. Public ratings remain a future candidate; their value and moderation costs need user and supervisor review. [S3], [T3], [K3], [A3]

6.5 A manageable exchange model

Keep payments and credit balances outside the first version. Simbi's credits address exchanges without an immediate reciprocal match, but introduce additional accounting rules. Test how often direct matches are possible and define what students can do when no suitable reciprocal partner is available. Confirm pilot eligibility separately. [S1], [S2], [K1], [T4]

Machine-learning and graph-based ranking remain future work under SSE-3; formal grading is outside the current scope. Calendar integrations and additional institutional administration need separate justification before being added.

Conclusion: existing services already address broad exchange, language practice, campus tutoring, and professional feedback. Our proposed contribution is a clear, multi-subject exchange journey for university students: describe both learning roles, discover a suitable partner, and agree on the exchange. Student research and usability tests must establish local demand, usefulness, and whether this journey improves on students' current alternatives.

7. Sources and access record

Every source below was accessed or retrieved on **23 October 2026**. Publication or update dates, where noted, are separate from the access date. Official sources describe their own products and may contain promotional language; performance statistics and testimonials were not treated as independent evidence.

ID	Official source	Access date	Evidence used and retrieval notes
S1	Simbi: How It Works	2026-10-23	Official indexed extract describes conversations, proposed exchanges, and direct/credit exchange. Full-page retrieval was unsuccessful, including a 403 response; detailed workflow states remain Unknown.
S2	Simbi: All Services	2026-10-23	Category browsing, service listings, request navigation, and the explanation of internal credits.

ID	Official source	Access date	Evidence used and retrieval notes
S3	Simbi: sampled public service listing	2026-10-23	Qualifications, availability notes, reviews, and communication described on one public listing. Later retrieval failed. The sample does not establish every member's practices or verify delivery of the service.
S4	Simbi Community Development: official organizational profile	2026-10-23	Free joining/cashless model, public requests, and review display.
T1	Tandem: product homepage	2026-10-23	Language exchange, partner discovery, communication modes, and application review.
T2	Tandem: Everything You Can Do on Tandem	2026-10-23	Profile fields, language-combination feed, free/Pro distinction, and described interaction tools. Exact local prices were not established.
T3	Tandem: Community Guidelines	2026-10-23	Authentic-profile rules, reporting, moderation, and appeals. Page states last updated 14 July 2025.
T4	Tandem for Universities	2026-10-23	University partnerships, rollout/onboarding support, premium access, and optional usage reporting. Campus-only access and partnership prices remain Unknown.
K1	Knack: Students	2026-10-23	Campus peer tutoring, communication formats, institution-funded access, and student/course coverage limitations.
K2	Knack: Completing your Tutor Profile	2026-10-23	Tutor biography, experience, and availability. Page dated 3 June 2026.
K3	Knack: Open Student Requests	2026-10-23	Course-based request feed, tutor responses, approval, and possible training requirements. Page dated 29 October 2024.
K4	Knack: Scheduling a Tutoring Session	2026-10-23	Session details, tutor responses, recurrence, and eligible Instant Booking. Page dated 3 June 2026.
A1	ADPList: product homepage	2026-10-23	Professional mentor profiles, goal-aligned suggestions, and one-to-one sessions.
A2	ADPList: Find a Mentor	2026-10-23	Expertise-based discovery, student audience, profiles/reviews, and the booking journey.
A3	ADPList: Become a Mentor	2026-10-23	Mentor application review, topics, profiles, availability, notes, reviews, and mentor/mentee roles. Page states updated 15 October 2026; free-booking wording is qualified by A4.
A4	ADPList: Best Mentorship Platforms in 2026	2026-10-23	Used only for ADPList's own free/paid-session description and variable availability. Page states updated 15 October 2026. Its comparisons of other providers were not used as evidence about them.
P1	Student Skill Exchange: repository README	2026-10-23	Public README inspected in the browser; repository latest commit displayed as `6234b14e69aa693151e0fe9e61dcbb54e2bdd77e`. Used to align proposals with the stated project scope.

Related team document

SSE-3 - Literature Review: Peer-to-Peer Skill Exchange for University Students, supplied on 23 October 2026. Sections 6 and 7 inform the scope decisions in this report. This is a team synthesis, distinct from the official product sources above.

8. Review and delivery checklist

- [x] Four relevant systems are compared using the same ten criteria.
- [x] The selection reasons are explained.
- [x] Product claims are linked to sources with access dates.
- [x] Unverified capabilities are marked Unknown.
- [x] Observations are separated from team interpretations and proposals.
- [x] The conclusion explains existing approaches and the intended project contribution.
- [ ] Nursultan has reviewed this document through a pull request.
- [ ] Review comments have been resolved and the changes merged into `main`.
- [ ] The pull request is linked to the corresponding YouTrack task.

Suggested branch: `SSE-4-existing-systems-comparison`. Commit this file at `docs/research/existing-systems-comparison.md` and open a pull request into `main` with a title such as `SSE-4: compare existing skill exchange platforms`. Include the SSE-4 task link in the PR description. Nursultan must review the document; resolve any comments before merging, then add the PR link to SSE-4. Mark the task complete only after these steps are finished.

[S1]: https://simbi.com/howitworks

[S2]: https://simbi.com/services

[S3]: https://simbi.com/jennifer-ferencz-barato/reserved-for-anat

[S4]: https://simbi.com/simbi

[T1]: https://tandem.net/

[T2]: https://tandem.net/blog/tandem-app-overview

[T3]: https://tandem.net/pages/community-guidelines

[T4]: https://tandem.net/pages/partnerships/universities

[K1]: https://www.joinknack.com/students

[K2]: https://help.joinknack.com/en/articles/3636944-completing-your-tutor-profile

[K3]: https://help.joinknack.com/en/articles/3636980-open-student-requests

[K4]: https://help.joinknack.com/en/articles/3636906-scheduling-a-tutoring-session

[A1]: https://adplist.org/

[A2]: https://adplist.org/find-a-mentor

[A3]: https://adplist.org/become-a-mentor

[A4]: https://adplist.org/guides/best-mentorship-platforms

[P1]: https://github.com/HashiraOfSun/student-skill-exchange

5. Technology comparison and selected stack

Source document: docs/architecture/technology-comparison.md

Project: Development of a Peer-to-Peer Skill Exchange Platform for University Students

Task: SSE-5

Status: Ready for team review

1. Decision context

The first version is a web application for two-sided skill exchange. It must support accounts, profiles, offered and wanted skills, search, reciprocal partner suggestions, exchange requests, and agreed exchange details. The project proposal excludes built-in chat, video calls, payments, native mobile applications, and machine-learning recommendations from the MVP.

The system will be developed by two students: one is responsible mainly for the backend and database, and the other for the frontend and API integration. The repository does not yet provide evidence that either team member has production experience with a specific framework. This comparison therefore does not claim undocumented expertise. Before implementation begins, both developers should complete a small setup exercise with the selected stack; if that reveals a serious skill gap, the decision should be reviewed.

The options are assessed against the same project needs:

- **Functional fit:** suitability for accounts, related skill data, search, matching, and request-state rules.
- **Maintainability:** clear structure, type safety, testability, documentation, and manageable dependencies.
- **Security:** support for authentication, authorization, validation, secure transport, and safe defaults.
- **Deployment:** ability to run locally and on common managed or container-based hosting.
- **Team fit:** learning burden, division between frontend and backend, and ease of collaboration.
- **MVP effort:** the amount of framework and infrastructure work required before product features can be implemented.

2. Frontend application

Alternatives

Option	Functional fit and maintainability	Security and deployment	Team fit and MVP effort	Main trade-off
React + TypeScript + Vite	React builds interfaces from reusable components. TypeScript can detect many type errors before execution. Vite provides a development server and produces static production assets. This is a direct fit for the planned profile, search, recommendation, and request screens.	The production output can be served as static files. Client-side code still requires safe token handling, output encoding, dependency updates, and server-side authorization; React alone is not a security boundary.	One frontend codebase and one language shared with the selected backend reduce context switching. The team still needs to learn routing, forms, API state, and testing.	The team must select and maintain supporting libraries because React is a UI library rather than a complete application framework.
Next.js + TypeScript	Next.js adds routing, server rendering, server components, and other full-stack features to React. It is useful when server rendering or a combined full-stack application is required.	It can run as a Node.js server, a container, or a static export, although static export has limited feature support. The larger execution model creates more deployment and security surface than this MVP needs.	Familiar React concepts can transfer, but the team must also understand server/client boundaries, caching, and framework conventions.	Its full-stack and rendering features overlap with the separately owned Express backend and add complexity without a current requirement for server-side rendering.
Vue + TypeScript + Vite	Vue is an approachable, component-based framework with an incrementally adoptable ecosystem. It can implement all planned MVP screens and can use the same Vite deployment model.	Like a React single-page application, it can be deployed as static assets and depends on backend authorization for protected operations.	Its template syntax may be approachable, but choosing it would create a separate learning path if the team is already using React-oriented materials or prototypes. No such experience is documented yet.	Technically suitable, but it offers no project-specific advantage strong enough to replace the selected React path.

Decision

Select **React with TypeScript and Vite**.

This option provides the component model needed for the planned interface while keeping deployment independent from the backend. TypeScript is selected for static checking at the frontend/backend contract boundary. Vite is selected because it provides a focused development server and a static production build without adding server-rendering features that the MVP does not require.

The trade-off is that routing, form validation, API-state handling, and testing are not supplied as one integrated framework. These libraries should be selected only when the relevant implementation task begins, rather than added speculatively.

Official documentation: React: Describing the UI, TypeScript: A Static Type Checker, Vite: Getting Started, and Vite: Building for Production.

3. Backend platform

Alternatives

Option	Functional fit and maintainability	Security and deployment	Team fit and MVP effort	Main trade-off
Node.js + Express + TypeScript	Express supplies HTTP routing and middleware with a small core. TypeScript supports shared request and response types and catches many mistakes before execution. The modular approach fits a small REST API with explicit service and data-access layers.	Express documents production practices such as TLS, input validation, secure headers, dependency maintenance, and brute-force protection. These controls must be configured by the team. A Node process is straightforward to run locally or package in a container.	Sharing TypeScript with the frontend simplifies collaboration and API type definitions. Express has a small initial surface, but the team must establish its own project structure and choose validation, authentication, and database libraries.	Minimalism gives control but fewer secure defaults; inconsistent middleware or architecture is a risk unless conventions and tests are enforced.

Option	Functional fit and maintainability	Security and deployment	Team fit and MVP effort	Main trade-off
Django	Django includes an ORM, authentication, forms, and administration features. Its integrated conventions can speed up data-oriented applications and reduce dependency selection.	Django documents built-in protection against several common web risks, including SQL injection, cross-site scripting, cross-site request forgery, and clickjacking when used correctly. It runs as a Python web application and typically needs an additional REST API layer.	It is productive when the backend developer already knows Python and Django, but that experience is not recorded. It would also introduce Python alongside the TypeScript frontend.	Strong built-in features, but some are unnecessary for a separate JSON API and the two-language stack adds context switching.
Java + Spring Boot	Spring Boot provides a structured ecosystem, dependency injection, validation, security integration, testing, and production features. It is well suited to large, long-lived services.	Official documentation covers executable applications, container images, external configuration, monitoring, and production deployment. Security is powerful but requires correct Spring Security configuration.	It is a strong option if the backend developer already has substantial Java/Spring experience. Without that evidence, configuration and framework concepts create the largest learning cost of the three options.	More structure and production facilities than the diploma MVP currently needs, with higher setup and learning overhead.

Decision

Select **Node.js with Express and TypeScript**.

The MVP is primarily an I/O-oriented HTTP service rather than a CPU-intensive system. Node.js uses an event loop to support non-blocking I/O, while Express supplies a small routing and middleware layer. Using TypeScript on both sides of the API reduces language switching and makes it possible to share or generate interface types.

This decision does not assume that Express is secure automatically. The backend must use supported releases, TLS in deployment, schema-based input validation, centralized error handling, secure response headers, rate limiting on authentication endpoints, password hashing, and authorization checks in the service layer. Protected operations must never rely on frontend checks.

The trade-off is that the team must define and enforce structure. The backend should therefore be organised into route, validation, controller, service, repository, and domain modules, with automated tests for authorization and request-state transitions.

Official documentation: Node.js event loop, Express overview, Express middleware, Express security practices, and TypeScript Handbook.

4. Data storage

The core model contains strongly related records: users, skills, offered skills, wanted skills, reciprocal matches, exchange requests, status history, and agreed exchange details. Uniqueness, foreign-key relationships, and valid request-state changes matter more than flexible document shapes.

Alternatives

Option	Functional fit and maintainability	Security and deployment	Team fit and MVP effort	Main trade-off
PostgreSQL	Relational tables, primary keys, foreign keys, unique constraints, check constraints, transactions, and indexes match the connected domain model. Database constraints can protect invariants even when application code contains a defect.	A separate database server requires credentials, restricted network access, backups, and migrations. Managed PostgreSQL and containers are widely available deployment models. Application queries must still be parameterised and database roles must use least privilege.	SQL and explicit relations make the data model visible to both developers and suitable for architecture diagrams. Initial setup is greater than SQLite but appropriate for a multi-user web application.	Requires operating a database service and maintaining migrations. Schema changes are more deliberate than in a flexible document database.

Option	Functional fit and maintainability	Security and deployment	Team fit and MVP effort	Main trade-off
MongoDB	Its flexible document model is useful when records vary significantly and data accessed together can be embedded. This project, however, has many cross-record relationships and lifecycle rules. MongoDB supports multi-document transactions, but its documentation notes additional cost and recommends effective schema design rather than using transactions as a substitute.	Managed and container deployment are available. Correct authorization, network restriction, indexes, validation, and backup configuration remain necessary.	JavaScript/TypeScript tooling is convenient, but the team would need to choose carefully between embedding and referencing for skills, requests, and users.	Flexibility can reduce early migration work, but denormalisation and cross-document consistency add avoidable design complexity for this relational domain.
SQLite	SQLite is a self-contained, serverless, transactional SQL database and is excellent for local prototypes and tests. It supports the necessary relational concepts.	Deployment is simple because data is stored in one local file. The official guidance notes that SQLite allows one writer at a time and recommends a client/server database when many concurrent writers or multiple application servers are required.	It has the lowest setup cost and is useful for isolated tests, but moving from a local file to production PostgreSQL creates environment differences.	Simple for a prototype, but less suitable as the main database for a deployed multi-user service and future horizontal scaling.

Decision

Select **PostgreSQL** as the application database.

The decisive factor is data integrity. A user can offer and request many skills, and an exchange request links two participants and moves through controlled statuses. PostgreSQL foreign keys, unique constraints, check constraints, and transactions support these rules directly. It also provides conventional indexes for search fields and relationship queries.

The trade-off is operational work: the team must manage migrations, environment-specific connection settings, backups for a deployed instance, and least-privilege credentials. SQLite may still be used for small isolated experiments only if the selected data-access library produces equivalent behaviour; integration tests should run against PostgreSQL to avoid database-specific surprises.

Official documentation: PostgreSQL overview, PostgreSQL constraints, MongoDB data modelling, MongoDB transactions, About SQLite, and Appropriate Uses for SQLite.

5. Frontend-backend communication

Alternatives

Option	Functional fit and maintainability	Security and deployment	Team fit and MVP effort	Main trade-off
REST-style HTTP API with JSON and OpenAPI	Resource-oriented endpoints map clearly to profiles, skills, recommendations, and exchange requests. Standard HTTP methods and status codes provide familiar semantics. An OpenAPI document can serve as the shared contract and support generated documentation or clients.	HTTPS protects data in transit. The API still requires authentication, object-level authorization, validation, CORS restrictions, response-size limits, and safe error messages. It uses ordinary HTTP infrastructure and is easy to deploy and inspect.	Clear endpoint ownership supports parallel frontend/backend work. Mock responses and an OpenAPI contract can be prepared before implementation.	Some screens may require several requests, and careless endpoint design can return too much or too little data. Contract changes must be managed explicitly.
GraphQL over HTTP	Clients request selected fields through a typed schema, which can reduce over-fetching and combine nested data access. A single graph can be useful for multiple diverse clients.	A single endpoint does not remove authorization needs. Field-level access, query depth or cost, introspection policy, and error handling add security and operational work.	Both developers must learn schema, resolver, client-cache, and query conventions. The MVP has one web client and a small, predictable data model, so the flexibility has limited current value.	Powerful query flexibility, but more implementation and security complexity than the MVP needs.

Option	Functional fit and maintainability	Security and deployment	Team fit and MVP effort	Main trade-off
WebSocket connection	WebSockets provide two-way interactive browser-server communication without polling and suit chat or live presence. The project proposal explicitly excludes real-time chat, and the planned request workflow does not require a continuously open connection.	Persistent connections require connection authentication, origin checks, reconnect logic, resource limits, and compatible proxy configuration. Deployment and testing are more involved than ordinary request-response HTTP.	Adds a second communication model for a feature outside the MVP.	Best for real-time events, but unsuitable as the primary CRUD interface and unjustified for the present scope.

Decision

Select a **REST-style HTTP API using JSON over HTTPS**, documented with **OpenAPI**.

The MVP operations map naturally to resources and state-changing commands. Standard HTTP semantics are sufficient, the interface can be tested independently, and an OpenAPI contract gives both team members a reviewable integration boundary. API endpoints should be versioned under `/api/v1` to make the initial contract explicit.

The trade-off is that the team must design task-oriented endpoints carefully, especially for reciprocal recommendations and request-state transitions. The API should not expose database tables directly. GraphQL can be reconsidered only if later clients need substantially different data shapes. WebSockets can be reconsidered only if validated requirements add real-time chat or notifications.

Official documentation and standards: RFC 9110: HTTP Semantics, OpenAPI Specification, GraphQL queries, GraphQL over HTTP, and MDN WebSocket API.

6. Selected stack

Layer	Selected technology	Reason for selection
Frontend	React + TypeScript + Vite	Component-based UI, static deployment, type checking, and no unnecessary server-rendering layer
Backend	Node.js + Express + TypeScript	Small HTTP API surface, shared language with frontend, and explicit modular architecture
Database	PostgreSQL	Relational integrity, constraints, transactions, and indexes for strongly connected domain data
Communication	REST-style JSON API over HTTPS	Simple request-response model for the MVP and broad tooling support
API contract	OpenAPI	Reviewable interface for parallel frontend/backend work and generated documentation

The initial deployment should consist of three independently configurable parts: static frontend assets, one backend service, and one PostgreSQL database. Local development may use containers or locally installed services, but secrets and database files must not be committed. Specific hosting providers are intentionally not selected here because cost, university requirements, and account availability still need confirmation.

7. Security and maintenance baseline

The technology choice does not by itself make the application secure. The implementation must include:

- supported dependency versions and automated dependency checks;
- HTTPS outside local development;
- password hashing with a maintained password-hashing library;
- server-side request validation and centralized error handling;
- authentication plus object-level authorization on every protected operation;
- parameterised database access and least-privilege database credentials;
- explicit CORS configuration for the deployed frontend origin;
- secure token or session handling agreed during the authentication design task;
- rate limiting for login and other abuse-sensitive endpoints;
- database migrations, backups for deployed data, and tested restore instructions;
- automated tests for access restrictions and exchange-request state transitions;
- an OpenAPI document kept consistent with the implemented API.

8. Consequences and review triggers

The selection favours a small, understandable architecture over framework breadth. It accepts additional responsibility for Express middleware, project structure, and security configuration in exchange for one language across the web stack and a clear separation of frontend and backend.

Review this decision if any of the following occurs:

- the team setup exercise shows a substantially stronger shared capability in Django or Spring Boot;
- university or supervisor requirements mandate another platform or deployment environment;
- validated requirements introduce real-time chat, multiple client applications, or complex live collaboration;
- the data model becomes predominantly unstructured rather than relational;
- deployment constraints make a separately hosted frontend, backend, and database impractical.

Otherwise, implementation and architecture tasks should use this document as the baseline.

6. Requirements

Source document: docs/requirements.md

Project: Development of a Peer-to-Peer Skill Exchange Platform for University Students

Task: SSE-6

Status: Initial requirements for team review

1. Purpose and scope

This document defines the initial functional and non-functional requirements for the first version of Student Skill Exchange. The first version supports the journey from registration to an agreed peer-learning exchange between two university students.

Built-in chat, video calls, payments, native mobile applications, machine-learning recommendations, certificates, formal skill assessment, group classes, and a course marketplace are outside the first-version scope.

2. Sources and validation status

Code	Source or status
P1	Project proposal: approved scope, objectives, main scenario, and engineering constraints.
R1	Literature review: rationale for transparent matching, user control, trust, and usable peer-learning interactions.
T1	Technology comparison: selected web stack, REST API, PostgreSQL, and security baseline.
Y1	YouTrack task SSE-6: required coverage and acceptance criteria.
A1	**Assumption - pending validation through SSE-2:** students are willing to create profiles and use reciprocal skill matching.
A2	**Assumption - pending validation through SSE-2:** email and password are acceptable credentials for the first version.
A3	**Assumption - pending validation through SSE-2:** the proposed profile fields, filters, and exchange-detail fields are sufficient.
A4	**Assumption - pending validation through SSE-2 and usability testing:** the usability and performance targets below are appropriate for target users.

Requirements marked **Scope-backed** come from the reviewed project scope. Requirements marked **Assumption** must be reviewed after SSE-2; they must not be presented as validated user findings.

Priorities follow MoSCoW:

- **Must:** required for the first-version main scenario.

- **Should:** valuable for the first version but not required to demonstrate the complete main scenario.
- **Could:** optional if time permits.

3. Functional requirements

3.1 Registration and authentication

FR-AUTH-01 - Register an account

- **Priority:** Must
- **Status:** Scope-backed; credential choice is an Assumption
- **Requirement:** A visitor shall be able to register using a unique email address, password, and display name.
- **Acceptance criteria:**

1. Valid unique details create one account and start an authenticated session or direct the user to sign in.
2. Missing or malformed fields are rejected with field-specific messages.
3. An email already assigned to an account is rejected without creating another account.

- **Source/rationale:** P1 main scenario and first-version scope; A2 for email/password choice.

FR-AUTH-02 - Sign in

- **Priority:** Must
- **Status:** Scope-backed; credential choice is an Assumption
- **Requirement:** A registered user shall be able to sign in with their email address and password.
- **Acceptance criteria:**

1. Correct credentials create an authenticated session and open the application.
2. Incorrect credentials do not create a session and produce a generic error that does not reveal whether the email exists.
3. An authenticated user can access protected functions until the session expires or is ended.

- **Source/rationale:** P1 main scenario and access-control scope; A2 for email/password choice.

FR-AUTH-03 - Sign out

- **Priority:** Must

- **Status:** Scope-backed
- **Requirement:** An authenticated user shall be able to sign out.
- **Acceptance criteria:** Signing out invalidates the current session and protected pages or API operations require authentication again.
- **Source/rationale:** P1 requirement for authentication and access control.

FR-AUTH-04 - Recover access

- **Priority:** Could
- **Status:** Assumption
- **Requirement:** A user may request a time-limited password-reset link for their registered email address.
- **Source/rationale:** Common account-recovery need; excluded from the mandatory main scenario until validated and scheduled.

3.2 Profiles

FR-PRO-01 - View and edit own profile

- **Priority:** Must
- **Status:** Scope-backed; exact fields are an Assumption
- **Requirement:** An authenticated user shall be able to view and edit their display name, short biography, and optional contact or study information approved for the first version.
- **Acceptance criteria:**

1. Saved valid changes appear when the profile is reopened.
2. Invalid values are rejected with field-specific messages.
3. A user cannot edit another user's profile through either the interface or direct API requests.

- **Source/rationale:** P1 profile scope and access-control challenge; A3 for exact fields.

FR-PRO-02 - View another student's exchange profile

- **Priority:** Must
- **Status:** Scope-backed; visible fields are an Assumption
- **Requirement:** An authenticated user shall be able to open another student's profile and see the information needed to assess a possible exchange, including offered and wanted skills.
- **Acceptance criteria:** The profile displays only fields designated as visible to other authenticated users and never exposes authentication data or private internal identifiers.
- **Source/rationale:** P1 search and partner-selection scenario; R1 rationale for informed and trustworthy partner selection; A3 for visible fields.

3.3 Skills

FR-SKL-01 - Manage offered skills

- **Priority:** Must
- **Status:** Scope-backed
- **Requirement:** A user shall be able to add and remove skills they can offer.
- **Acceptance criteria:**

1. Added skills appear on the user's profile and are available to search and matching.
2. Removing a skill removes it from future search and matching results without changing historical exchange records.
3. The same offered skill cannot be added twice to one profile.

- **Source/rationale:** P1 proposed solution and first-version scope.

FR-SKL-02 - Manage wanted skills

- **Priority:** Must
- **Status:** Scope-backed
- **Requirement:** A user shall be able to add and remove skills they want to learn.
- **Acceptance criteria:**

1. Added skills appear on the user's profile and are available to matching.
2. Removing a skill removes it from future matching results without changing historical exchange records.
3. The same wanted skill cannot be added twice to one profile.

- **Source/rationale:** P1 proposed solution and first-version scope.

FR-SKL-03 - Use a consistent skill catalogue

- **Priority:** Must
- **Status:** Scope-backed; catalogue contents are an Assumption
- **Requirement:** Offered and wanted skills shall reference a shared skill catalogue so that equivalent skills can be searched and matched consistently.
- **Acceptance criteria:** Selecting the same catalogue skill on two different profiles stores the same skill identifier, and unsupported duplicate catalogue entries cannot be created through the public API.
- **Source/rationale:** P1 requirement to handle repeated skills and match complementary profiles; A3 for initial catalogue contents.

3.4 Search

FR-SRCH-01 - Search and filter students

- **Priority:** Must
- **Status:** Scope-backed
- **Requirement:** An authenticated user shall be able to search for other students by offered skill and filter results by skill category.
- **Acceptance criteria:**

1. Results contain only profiles that offer the selected skill and satisfy the selected category filter.
2. The current user's own profile is not returned as a potential partner.
3. Clearing the query or filters returns the documented default result state.

- **Source/rationale:** P1 first-version scope and main scenario.

FR-SRCH-02 - Explain empty search results

- **Priority:** Must
- **Status:** Scope-backed
- **Requirement:** When no profiles match a search, the system shall display an empty state and allow the user to change or clear the filters.
- **Acceptance criteria:** A zero-result query shows no profile cards, displays a clear message, and provides an action that returns to a broader search.
- **Source/rationale:** P1 engineering challenge requiring clear handling when no suitable partner exists.

3.5 Partner matching

FR-MAT-01 - Generate reciprocal partner suggestions

- **Priority:** Must
- **Status:** Scope-backed; user value is an Assumption
- **Requirement:** The system shall suggest another user when the current user offers at least one skill the other user wants and the other user offers at least one skill the current user wants.
- **Acceptance criteria:**

1. Every suggested pair satisfies both directions of the reciprocal rule using active offered and wanted skills.
2. The current user is never suggested to themselves.

3. Removing one side of the only reciprocal skill pair removes that suggestion after matching is recalculated.

- **Source/rationale:** P1 proposed solution and objectives; A1 for whether students value reciprocal matching.

FR-MAT-02 - Explain each suggestion

- **Priority:** Must
- **Status:** Scope-backed
- **Requirement:** Each suggested partner shall show the offered and wanted skills that caused the reciprocal match.
- **Acceptance criteria:** For every suggestion, the interface displays at least one skill in each matching direction, and both displayed skills correspond to current profile data.
- **Source/rationale:** P1 first-version scope; R1 rationale for transparent, user-controllable matching.

FR-MAT-03 - Handle incomplete profiles and no matches

- **Priority:** Must
- **Status:** Scope-backed
- **Requirement:** The system shall provide clear guidance when matching cannot run because skills are missing or when no reciprocal match exists.
- **Acceptance criteria:**

1. A user with no offered or no wanted skill is prompted to complete the missing profile section.
2. A completed profile with no reciprocal matches receives a distinct no-matches message and can continue to manual search.

- **Source/rationale:** P1 engineering challenges.

3.6 Exchange requests

FR-REQ-01 - Send an exchange request

- **Priority:** Must
- **Status:** Scope-backed
- **Requirement:** An authenticated user shall be able to send another student an exchange request containing the proposed offered skill, wanted skill, and an optional short note.
- **Acceptance criteria:**

1. A valid request creates one record with status `Pending` and identifies the sender and recipient.
2. A user cannot send a request to themselves.
3. A request cannot use a skill not present in the relevant participant's active profile data.

4. A duplicate pending request for the same participants and skill pair is rejected.

- **Source/rationale:** P1 main user scenario and exchange-request scope.

FR-REQ-02 - View incoming and outgoing requests

- **Priority:** Must
- **Status:** Scope-backed
- **Requirement:** A user shall be able to view their incoming and outgoing requests with participant, skill pair, current status, and creation time.
- **Acceptance criteria:** The two lists show only requests in which the authenticated user is the recipient or sender respectively, and their displayed status matches stored data.
- **Source/rationale:** P1 first-version scope.

FR-REQ-03 - Accept or decline a request

- **Priority:** Must
- **Status:** Scope-backed
- **Requirement:** The recipient of a pending request shall be able to accept or decline it.
- **Acceptance criteria:**

1. Only the recipient can accept or decline the request.
2. A pending request changes exactly once to `Accepted` or `Declined`.
3. Repeating the operation or applying it to a non-pending request is rejected without changing stored data.

- **Source/rationale:** P1 proposed solution and request-state engineering challenge.

FR-REQ-04 - Cancel a pending request

- **Priority:** Must
- **Status:** Scope-backed; transition rule defined here
- **Requirement:** The sender shall be able to cancel their request while it is pending.
- **Acceptance criteria:**

1. Only the sender can cancel a pending request.
2. Cancellation changes the status to `Cancelled`.
3. Accepted, declined, or already cancelled requests cannot be cancelled again.

- **Source/rationale:** P1 states that requests can be cancelled according to rules defined in the requirements.

FR-REQ-05 - Preserve request status history

- **Priority:** Must
- **Status:** Scope-backed
- **Requirement:** The system shall record each request status change with the new status, responsible user, and timestamp.
- **Acceptance criteria:** Creating, accepting, declining, or cancelling a request creates a corresponding immutable history entry, and only the two participants can view that history.
- **Source/rationale:** P1 first-version scope and requirement for consistent request information.

3.7 Agreed exchange details

FR-AGR-01 - Record agreed details

- **Priority:** Must
- **Status:** Scope-backed; exact fields are an Assumption
- **Requirement:** After a request is accepted, either participant shall be able to record the agreed topic, date and time, and format (`In person` or `External online service`), with an optional location or meeting link as appropriate.
- **Acceptance criteria:**

1. Details can be created only for an accepted request and by one of its participants.
2. Topic, future date and time, and format are required.
3. The location or meeting-link field is validated according to the selected format.
4. Saved details are visible to both participants.

- **Source/rationale:** P1 proposed solution and main scenario; A3 for exact fields.

FR-AGR-02 - Update agreed details

- **Priority:** Must
- **Status:** Scope-backed; collaboration rule is an Assumption
- **Requirement:** Either participant shall be able to update the agreed details, and the system shall show when they were last updated and by whom.
- **Acceptance criteria:**

1. Only the two participants can update the details.
2. Valid changes are visible to both participants after saving.

3. The last-updated timestamp and responsible participant change with every successful update.

- **Source/rationale:** P1 requirement that both participants see consistent agreed details; A3 for the update workflow.

FR-AGR-03 - Keep sessions outside the platform

- **Priority:** Must
- **Status:** Scope-backed
- **Requirement:** The first version shall direct users to conduct the learning session in person or through an external service and shall not provide built-in chat or video calls.
- **Acceptance criteria:** No built-in chat or video-call function appears in the first-version navigation or API, while the agreed format and optional external meeting information can be recorded.
- **Source/rationale:** P1 explicit scope exclusions.

4. Non-functional requirements

4.1 Security and privacy

NFR-SEC-01 - Protect passwords

- **Priority:** Must
- **Status:** Scope-backed
- **Requirement:** Passwords shall be processed only by the backend and stored using a maintained adaptive password-hashing algorithm with a unique salt.
- **Acceptance criteria:** Database inspection shows no plaintext or reversibly encrypted passwords; automated authentication tests confirm that password verification succeeds only for the correct password.
- **Source/rationale:** P1 authentication and access-control scope; T1 security baseline.

NFR-SEC-02 - Enforce authentication and object-level authorization

- **Priority:** Must
- **Status:** Scope-backed
- **Requirement:** Every protected API operation shall verify authentication and permission for the specific profile, request, or agreement being accessed.
- **Acceptance criteria:** Automated tests cover every protected endpoint: unauthenticated access returns `401`, and authenticated cross-user access outside the permitted relationship returns `403` or a non-disclosing `404`.
- **Source/rationale:** P1 engineering challenges; T1 security baseline.

NFR-SEC-03 - Validate untrusted input

- **Priority:** Must
- **Status:** Scope-backed
- **Requirement:** The backend shall validate all request parameters and bodies against defined schemas and use parameterised database access.
- **Acceptance criteria:** Invalid type, length, format, and unexpected-field tests are rejected with `4xx` responses; automated security tests confirm common SQL-injection strings are treated as data and do not alter query structure.
- **Source/rationale:** T1 selected REST/PostgreSQL security responsibilities.

NFR-SEC-04 - Protect transport and secrets

- **Priority:** Must
- **Status:** Scope-backed
- **Requirement:** Non-local deployments shall use HTTPS, and credentials or signing secrets shall be supplied through protected environment configuration rather than committed files.
- **Acceptance criteria:** The deployed HTTP endpoint redirects to HTTPS or is unavailable; repository secret scanning finds no active credentials; production cookies, if used, have `Secure`, `HttpOnly`, and an explicit `SameSite` policy.
- **Source/rationale:** T1 security and deployment baseline.

NFR-SEC-05 - Limit authentication abuse

- **Priority:** Must
- **Status:** Assumption
- **Requirement:** Login attempts shall be rate-limited to no more than 10 failed attempts for the same account or client address within 15 minutes.
- **Acceptance criteria:** An automated test shows that attempts beyond the configured threshold receive `429 Too Many Requests` and that access becomes available after the limit window expires.
- **Source/rationale:** T1 Express security baseline; threshold requires review after testing.

NFR-SEC-06 - Minimise exposed personal data

- **Priority:** Must
- **Status:** Scope-backed; exact visible fields are an Assumption
- **Requirement:** API responses and screens shall expose only the profile fields required for finding and arranging an exchange.
- **Acceptance criteria:** Public profile responses exclude password hashes, session data, private configuration, and email unless the visibility decision explicitly permits it; API contract tests verify the approved response schema.

- **Source/rationale:** P1 target-user scenario; R1 trust rationale; A3 for final visible fields.

4.2 Usability and accessibility

NFR-USA-01 - Complete the main scenario

- **Priority:** Must
- **Status:** Assumption
- **Requirement:** At least 80% of five representative university-student test participants shall complete registration, skill setup, partner discovery, request submission, request acceptance, and recording agreed details without moderator assistance.
- **Acceptance criteria:** A documented usability test records at least four successful completions out of five participants and lists failures or required corrections.
- **Source/rationale:** P1 main scenario and evaluation objective; A4 target pending user research and evaluation planning.

NFR-USA-02 - Provide clear system feedback

- **Priority:** Must
- **Status:** Scope-backed
- **Requirement:** Forms and asynchronous actions shall show validation errors, loading state, success confirmation, and recoverable error guidance where applicable.
- **Acceptance criteria:** A UI review of registration, login, profile, search, request, and agreement screens confirms that each applicable state is represented and that errors identify the affected action or field.
- **Source/rationale:** P1 frontend responsibilities and requirement to communicate incomplete/no-match cases clearly.

NFR-USA-03 - Support keyboard use and readable labels

- **Priority:** Must
- **Status:** Assumption
- **Requirement:** Core screens shall support keyboard navigation, visible focus, programmatic form labels, and text/background contrast conforming to WCAG 2.2 Level AA for the tested components.
- **Acceptance criteria:** Automated accessibility checks report no critical violations on core screens, and a manual keyboard test completes the main scenario without a mouse.
- **Source/rationale:** A4 usability assumption; inclusive access for the target student population.

4.3 Performance

NFR-PERF-01 - API response time

- **Priority:** Must
- **Status:** Assumption
- **Requirement:** Under 20 concurrent simulated users with a seeded dataset of at least 1,000 users and 10,000 user-skill relationships, the 95th-percentile response time shall be at most 500 ms for profile and request operations and at most 1,000 ms for search and reciprocal matching.
- **Acceptance criteria:** A repeatable load-test report from the documented test environment meets both 95th-percentile targets with an API error rate below 1%.
- **Source/rationale:** A4 initial measurable target; P1 requires usable search and matching.

NFR-PERF-02 - Core page loading

- **Priority:** Should
- **Status:** Assumption
- **Requirement:** On the agreed staging environment, each core page should achieve a Largest Contentful Paint of at most 2.5 seconds in a documented Lighthouse mobile test.
- **Source/rationale:** A4 initial usability/performance target; selected static frontend in T1.

4.4 Compatibility and responsive design

NFR-COM-01 - Browser compatibility

- **Priority:** Must
- **Status:** Assumption
- **Requirement:** The main scenario shall work in the latest two stable major versions available at test time of Chrome, Firefox, Edge, and Safari.
- **Acceptance criteria:** A compatibility matrix records a passing main-scenario test for every listed browser/version combination, or records and resolves any blocking difference before release.
- **Source/rationale:** A4 compatibility target for a browser-based university platform.

NFR-COM-02 - Responsive web interface

- **Priority:** Must
- **Status:** Scope-backed; dimensions are an Assumption
- **Requirement:** Core screens shall remain usable at viewport widths from 360 px to 1440 px without unintended horizontal scrolling or hidden primary actions.
- **Acceptance criteria:** Registration, login, profile, search, recommendations, requests, and agreement screens pass manual checks at 360, 768, 1024, and 1440 px widths.
- **Source/rationale:** P1 requires a web interface adapted for desktop and mobile screens; A4 for chosen test widths.

4.5 Reliability and maintainability

NFR-REL-01 - Preserve request consistency

- **Priority:** Must
- **Status:** Scope-backed
- **Requirement:** Exchange-request status changes and their history entries shall be committed atomically.
- **Acceptance criteria:** Integration tests show that an induced failure cannot leave a changed request status without its history entry, or a history entry without the corresponding status change.
- **Source/rationale:** P1 consistency challenge; PostgreSQL transaction decision in T1.

NFR-MNT-01 - Maintain the API contract

- **Priority:** Must
- **Status:** Scope-backed
- **Requirement:** The implemented HTTP API shall have a versioned OpenAPI document consistent with its request schemas, responses, authentication rules, and error codes.
- **Acceptance criteria:** The OpenAPI document passes schema validation, and automated contract tests fail when the implementation no longer matches the documented endpoints used by the main scenario.
- **Source/rationale:** P1 expected API documentation; T1 selected OpenAPI contract.

NFR-MNT-02 - Verify mandatory requirements

- **Priority:** Must
- **Status:** Scope-backed
- **Requirement:** Every Must requirement in this document shall be traceable to at least one automated test, manual test case, or review record before the first-version evaluation.
- **Acceptance criteria:** A traceability table lists every Must requirement ID with evidence and contains no empty evidence field at release review.
- **Source/rationale:** Y1 requires clear, verifiable acceptance criteria; P1 includes testing and evaluation objectives.

5. Business rules and state model

5.1 Exchange-request states

Current state	Allowed action	Actor	New state
No request	Send valid request	Sender	Pending
Pending	Accept	Recipient	Accepted
Pending	Decline	Recipient	Declined

Current state	Allowed action	Actor	New state
Pending	Cancel	Sender	Cancelled
Accepted	Record or update agreement	Either participant	Accepted
Declined	None	-	Declined
Cancelled	None	-	Cancelled

Terminal requests are retained for history. A new request for the same users and skill pair may be considered only after the previous request is no longer pending; whether a waiting period is needed remains an assumption for later validation.

5.2 Matching rule

For current user A, user B is a reciprocal match when:

1. A and B are different active users;
2. at least one active skill offered by A is wanted by B; and
3. at least one active skill offered by B is wanted by A.

The interface must show the skills satisfying conditions 2 and 3. Ranking beyond the existence of a reciprocal match is outside the mandatory first-version requirement and may be defined after user research.

6. Traceability summary

Required area from SSE-6	Requirement IDs
Registration and authentication	FR-AUTH-01-04
Profiles	FR-PRO-01-02
Skills	FR-SKL-01-03
Search	FR-SRCH-01-02
Partner matching	FR-MAT-01-03
Exchange requests	FR-REQ-01-05
Agreed exchange details	FR-AGR-01-03
Security	NFR-SEC-01-06
Usability	NFR-USA-01-03
Performance	NFR-PERF-01-02
Compatibility	NFR-COM-01-02
Reliability and maintainability	NFR-REL-01, NFR-MNT-01-02

7. Assumption review after SSE-2

After the student research task is completed, the team shall review A1-A4 and record one of the following for every affected requirement:

- **Validated:** supported by collected evidence;
- **Revised:** changed to reflect collected evidence; or
- **Rejected:** removed from the planned scope with a recorded reason.

The review should specifically check credential preferences, profile information, skill terminology, useful search filters, willingness to use reciprocal matching, exchange-request expectations, agreement fields, and the practicality of the usability targets.

8. Team agreement

Agreement will be recorded through review and approval of the pull request for SSE-6.

- [ ] Nursultan Makhmutov confirms that the requirements fit the first-version scope.
- [ ] Tamir Burlibaev confirms that the requirements fit the first-version scope.

The approved pull request URL will be added to SSE-6.

7. System architecture

Source document: docs/architecture/system-architecture.md

Project: Development of a Peer-to-Peer Skill Exchange Platform for University Students

Task: SSE-7

Status: Draft for team review

Owner: Nursultan Makhmutov

Required reviewer: Tamir Burlibaev

1. Purpose and scope

This document describes the initial architecture of the Student Skill Exchange MVP. It aligns the selected technology stack with the functional and non-functional requirements for accounts, profiles, skills, search, reciprocal matching, exchange requests, and agreed exchange details.

The architecture deliberately excludes built-in chat, video calls, payments, native mobile applications, and machine-learning recommendations. Students conduct an agreed exchange in person or through an external service.

2. Architecture drivers

The design is guided by the following requirements:

- keep the frontend and backend independently deployable;
- provide a clear REST-style JSON API documented with OpenAPI;
- enforce authentication, object-level authorization, validation, and safe error handling on the backend;
- preserve relational integrity and request-state history with PostgreSQL constraints and transactions;
- support transparent reciprocal matching based on offered and wanted skills;
- keep request and agreement workflows consistent for both participants;
- remain small enough for a two-student team to implement and test.

3. System context

Editable source: `system-context.drawio`.

University students use the platform in two complementary roles: seeking help and offering help. A user can hold both roles at the same time. The platform supports finding a partner and agreeing on an exchange. The learning session itself takes place outside the platform, either in person or through an external service selected by the participants.

4. Use cases

Editable source: `use-case.drawio`.

The main end-to-end scenario is:

1. A visitor registers and logs in.
2. The student completes a profile and records offered and wanted skills.
3. The student searches manually or reviews reciprocal partner suggestions.
4. The student opens a partner profile and sees why the pair matches.
5. The student sends an exchange request.
6. The recipient accepts or declines the request.
7. After acceptance, either participant records the agreed topic, date, time, and format.

The sender can cancel a request only while it is pending. Built-in messaging and video sessions are outside the MVP.

5. Component architecture

Editable source: `component-diagram.drawio`.

5.1 Frontend

The React application is responsible for pages, forms, routing, client-side feedback, and calling the backend API. Client-side checks improve usability but do not replace backend authorization or validation.

5.2 Backend

The Express application exposes the REST API and separates transport logic from domain behaviour:

- **Routes and controllers** translate HTTP requests and responses.
- **Security middleware** authenticates requests, checks authorization, validates input, and rate-limits authentication attempts.

- **Authentication service** manages registration, login, logout, and password verification.
- **Profile and skill service** manages profile fields and offered/wanted skills.
- **Search and matching service** performs catalogue search and reciprocal matching.
- **Exchange request service** enforces request ownership and allowed status transitions.
- **Agreement service** manages the topic, time, format, and optional meeting information for accepted requests.
- **Repository layer** owns database access and transaction boundaries.

5.3 Database

PostgreSQL stores users, profiles, the skill catalogue, offered and wanted skills, exchange requests, immutable status-history records, and agreements. Foreign keys, unique constraints, check constraints, and transactions protect the principal invariants.

5.4 API contract

OpenAPI is the shared contract between the frontend and backend. It documents request and response schemas, authentication, validation errors, and status codes. The implemented API and the contract must be checked together.

6. Principal data relationships

- One user has one profile.
- A user can offer many skills and want many skills.
- Catalogue skills are reused across profiles rather than stored as free-text duplicates.
- An exchange request connects one sender, one recipient, one offered skill, and one wanted skill.
- A request has one current status and many immutable status-history records.
- An accepted request may have one agreement visible to both participants.

A detailed entity-relationship diagram can be added when the database-schema task begins. It is not required to replace the three architectural diagrams in this task.

7. Key runtime flows

7.1 Reciprocal matching

1. The frontend requests partner suggestions for the authenticated user.
2. The backend verifies the session and loads active offered and wanted skills.

3. The matching service selects users for whom both directions of the skill relationship exist.
4. The response includes the skills that explain each suggestion.

7.2 Exchange request

1. The sender selects the offered and wanted skills and submits a request.
2. The backend validates both skills against current profile data and rejects self-requests or duplicate pending requests.
3. The request and its initial history entry are committed in one database transaction.
4. Only the recipient can accept or decline; only the sender can cancel while the request is pending.

7.3 Agreement

1. Either participant opens an accepted request.
2. The backend verifies participation and accepted status.
3. The participant records or updates the topic, future date and time, format, and optional location or external link.
4. Both participants receive the same stored agreement details.

8. Security and privacy responsibilities

- Passwords are hashed on the backend with a maintained adaptive algorithm and unique salts.
- Every protected operation verifies authentication and permission for the specific resource.
- Request bodies and parameters are schema-validated before domain processing.
- Database operations use parameterised queries.
- Authentication endpoints are rate-limited.
- Secrets are provided through protected environment configuration and are not committed.
- Non-local traffic uses HTTPS.
- API responses expose only fields required for finding and arranging an exchange.
- The platform does not publish private contact information by default.

9. Deployment view

The initial deployment consists of three independently configurable parts:

1. static React production assets served through an HTTPS-capable web host;
2. a Node.js/Express API service configured through environment variables;
3. a PostgreSQL database accessible only to the API service.

The frontend receives only the public API base URL. Database credentials and authentication secrets remain on the backend. A reverse proxy or managed host terminates HTTPS and forwards API traffic to Express.

10. Requirement-to-component traceability

Requirement area	Primary components
Registration and authentication	Frontend forms, security middleware, authentication service, user repository
Profiles and skills	Profile UI, profile and skill service, profile/skill repositories
Search	Search UI, search service, indexed skill relationships
Reciprocal matching	Match suggestions UI, matching service, user-skill relationships
Exchange requests	Request UI, request service, request and history repositories
Agreed exchange details	Agreement UI, agreement service, agreement repository
Security and privacy	Security middleware, all domain services, environment configuration
API documentation	OpenAPI contract and contract checks

11. Assumptions and planned refinement

The component diagram is intentionally a draft. Service boundaries may be combined in code if the implementation remains clear and testable. Exact deployment providers, authentication-session mechanism, database-access library, and supporting frontend libraries will be selected during implementation tasks.

User-research findings must be used to review assumptions about profile fields, search filters, matching value, agreement fields, privacy, and meeting format. Changes must update the requirements, diagrams, and OpenAPI contract together.

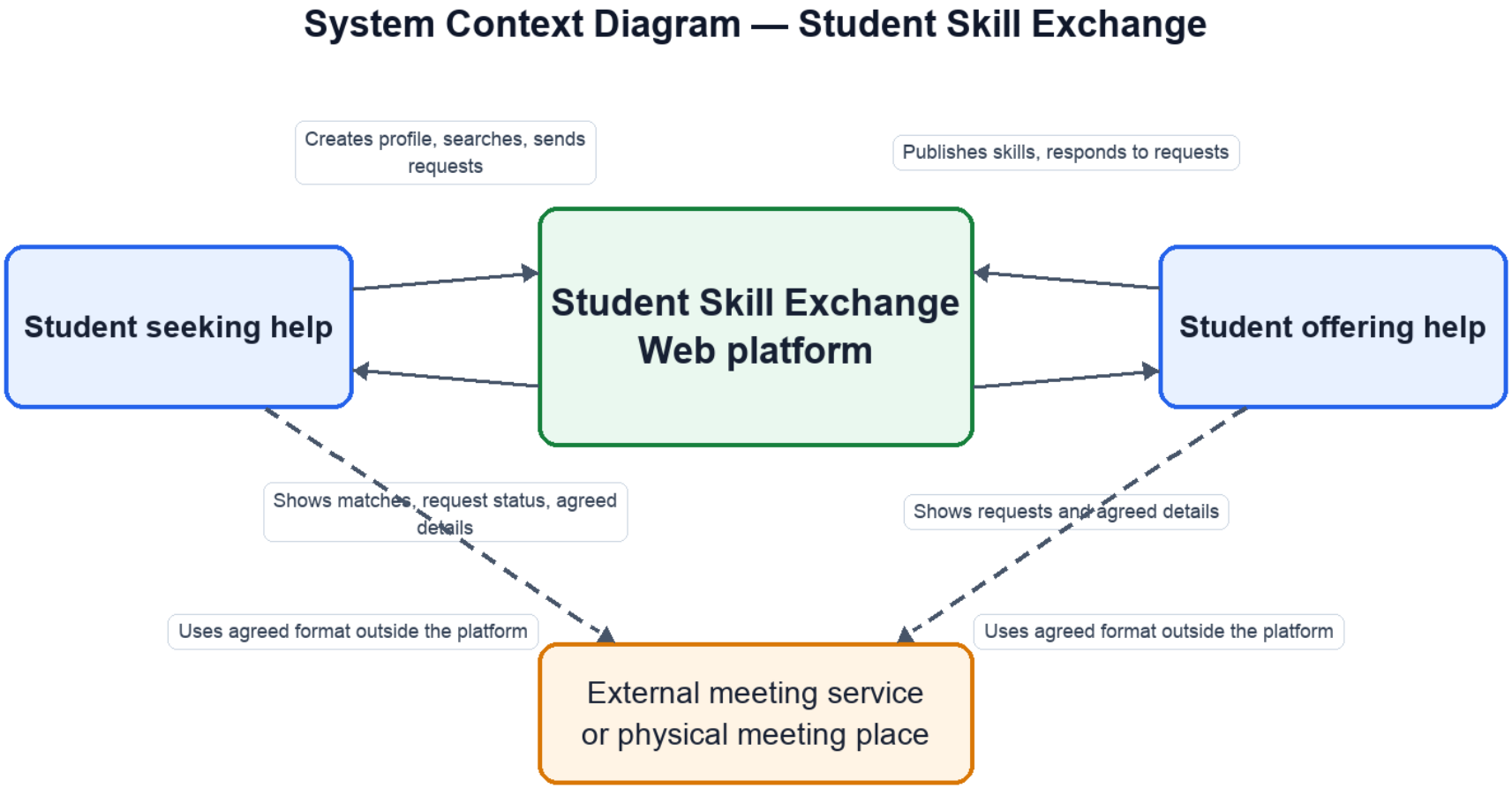
12. Review and delivery checklist

- [x] System context diagram is included.
- [x] Use case diagram is included.
- [x] Draft component diagram is included.
- [x] Editable Draw.io sources and readable PNG exports are included.
- [x] Diagram elements align with the selected stack and current MVP scope.

- [x] Requirements, components, data, and main scenarios are traceable.
- [ ] Tamir has reviewed the document through a pull request.
- [ ] Review comments have been resolved and the changes merged into `main`.
- [ ] The pull request is linked to SSE-7.

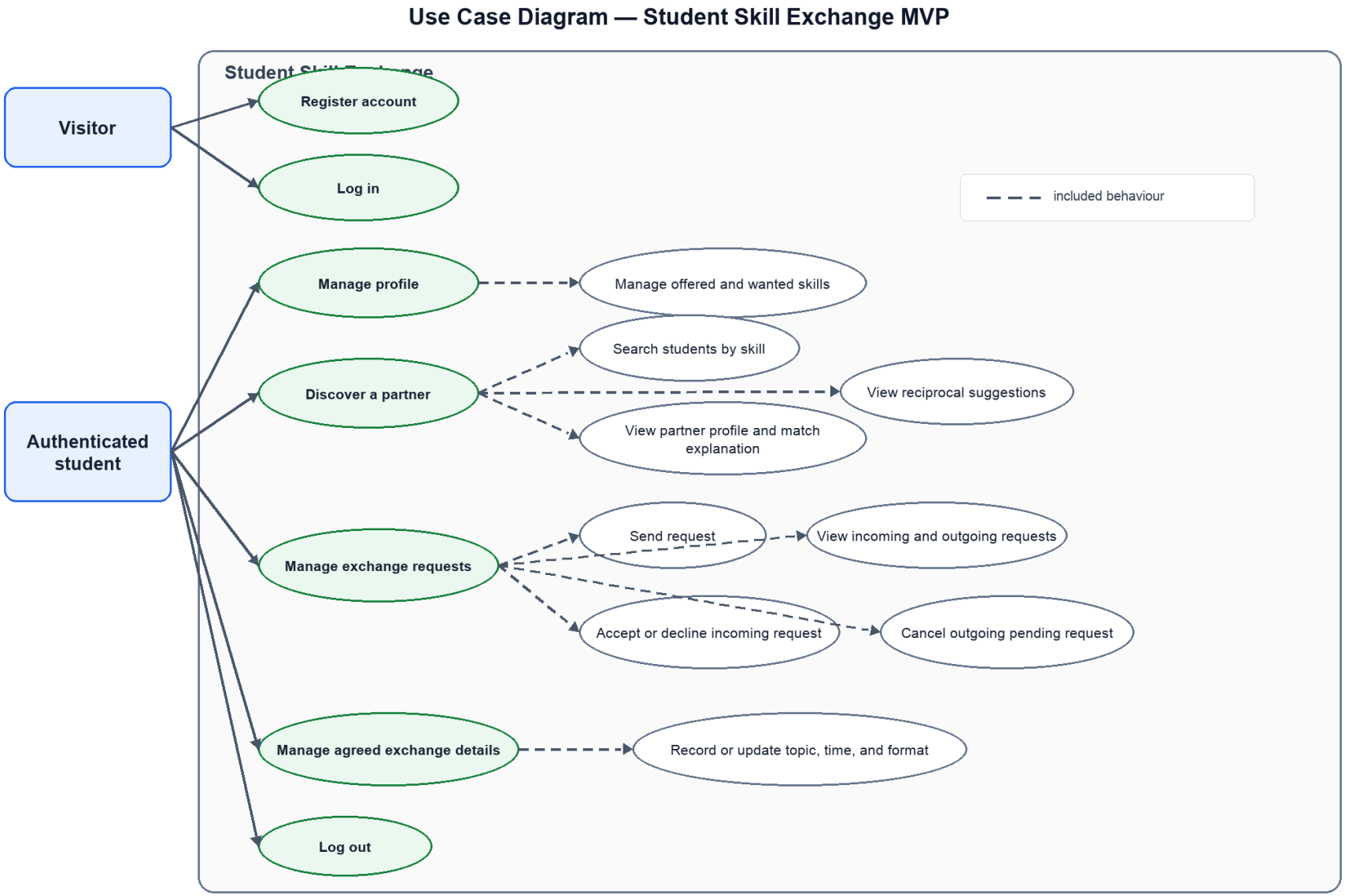
Suggested branch: `SSE-7-system-architecture`.

System context diagram



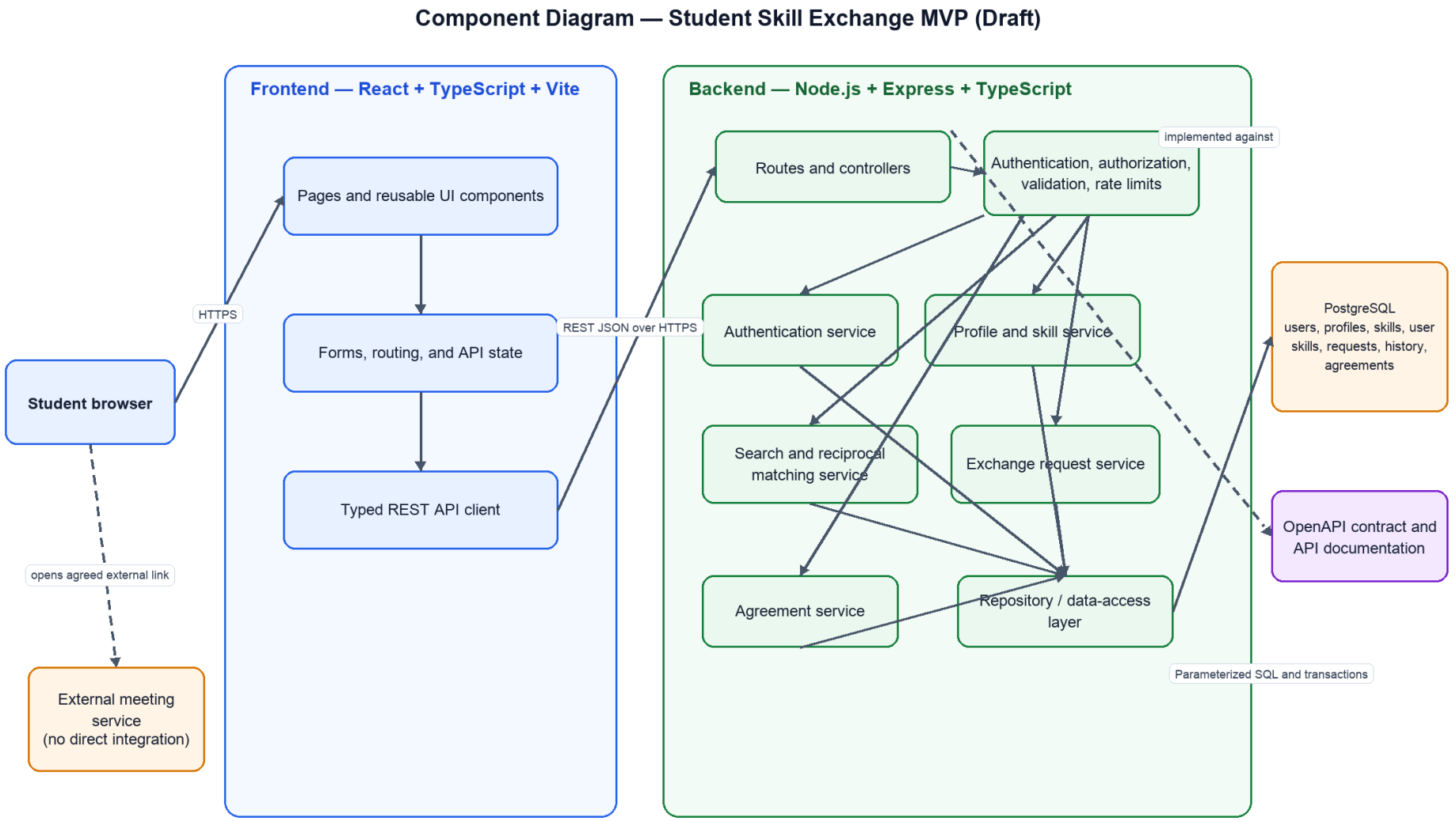
System context diagram

Use case diagram



Use case diagram

Component diagram



Component diagram

8. Interactive UI prototype

Source artifact: SSE-8_UI_Prototype.zip

The repository includes a self-contained interactive HTML prototype for the core Student Skill Exchange journey. It demonstrates the proposed interaction model before implementation and keeps sample data inside the browser.

Prototype area	Demonstrated behavior
Account and profile	Sample sign-in/registration, profile details, offered skills, and wanted skills.
Discovery	Search and category filtering across student profiles.
Reciprocal matching	Shows both directions of a possible exchange: what each participant offers and wants.
Exchange requests	Create, accept, decline, cancel, and review incoming or outgoing requests.
Shared plan	Record an accepted exchange's topic, Astana date/time, and in-person or external online format.
Validation and states	Empty, loading, error, duplicate-request, authorization, and status-history behaviors.
Accessibility	Keyboard focus, labels, error messages, responsive layouts, and visible interaction states.

Prototype limitation: the interface is a demonstrator. Data persistence, production authentication, backend APIs, database integration, deployment, and completed verification remain future implementation work.

9. Verification and evaluation plan

Source document: docs/testing/verification-and-evaluation-plan.md

Project: Development of a Peer-to-Peer Skill Exchange Platform for University Students

Task: SSE-9

Status: Planned; no tests in this document have been executed

Owner: Nursultan Makhmutov

Required reviewer: Tamir Burlibaev

1. Purpose

This document defines how the Student Skill Exchange MVP will be verified against its mandatory requirements and how the team will evaluate whether reciprocal matching helps university students find a suitable learning partner.

The plan is based on the initial platform requirements and the system architecture. It covers the first-version journey from registration to recorded exchange details. Built-in chat, video calls, payments, native mobile applications, machine-learning recommendations, certificates, and formal skill assessment remain outside scope.

2. Test status and evidence policy

Everything described here is **planned**. No pass, fail, performance, accessibility, compatibility, or usability result is claimed yet.

During implementation and evaluation, each execution record shall contain:

- test-case ID and requirement ID;
- application version or commit SHA;
- date, tester, environment, browser, and viewport where relevant;
- test data used;
- expected and actual result;
- pass, fail, blocked, or not-run status;
- defect link and retest evidence when applicable.

Automated reports, screenshots, compatibility matrices, usability notes, and defect links will be stored under `docs/testing/evidence/` or linked from the final report. Personal participant data shall not be committed.

3. Verification strategy

Level	Purpose	Planned evidence
Unit tests	Verify matching rules, validation, and request-state rules in isolation	Test-run report with commit SHA
API integration tests	Verify database behaviour, authorization, transactions, and HTTP responses	API test report and seeded database version
Contract tests	Verify implementation against OpenAPI schemas, authentication rules, and error codes	OpenAPI validation and contract-test report
End-to-end tests	Verify successful and unsuccessful user journeys through the browser	Browser test report, screenshots for failures
Security review	Verify password handling, input validation, access control, secrets, and transport settings	Security checklist and automated test output
Performance tests	Verify API response targets with the required concurrency and dataset	Repeatable load-test script, configuration, and report
Accessibility and UI review	Verify keyboard access, labels, focus, contrast, responsive layout, and feedback states	Automated scan plus manual checklists
Compatibility tests	Verify the main scenario in the required browser/version combinations	Compatibility matrix
Usability evaluation	Measure task completion, assistance, time, errors, comprehension, and participant feedback	Anonymised observation sheet and summary

4. Environments and test data

4.1 Environments

Environment	Use	Required configuration
Local automated test	Unit, integration, and contract tests	Isolated PostgreSQL test database; fixed clock where needed; test-only secrets
Staging	End-to-end, accessibility, responsive, compatibility, and usability tests	Production-like build over HTTPS; no production credentials or personal data
Load-test environment	Performance verification	Documented CPU, memory, database version, API version, and network location; no other workloads

The exact versions of Node.js, PostgreSQL, browsers, test libraries, and operating systems shall be recorded when the tests are run.

4.2 Seeded data

The functional dataset shall include at least:

- six active student accounts and one unauthenticated visitor;
- one account with an incomplete skill profile;
- one completed profile with no reciprocal matches;

- one reciprocal pair with exactly one skill in each direction;
- one profile with multiple offered and wanted skills;
- skill categories and enough catalogue entries to test search filters;
- pending, accepted, declined, and cancelled requests;
- one accepted request with agreement details;
- attempted duplicate, self-directed, and unauthorized operations.

The load-test dataset shall contain at least 1,000 synthetic users and 10,000 active user-skill relationships. Synthetic accounts shall use non-routable addresses such as `example.test`; no real student records are required.

4.3 Data reset and repeatability

Automated suites shall create or restore a known database state before execution. Test cases must not depend on execution order unless the sequence is explicitly part of a state-transition test. Failed runs must retain enough logs to reproduce the failure without retaining passwords, session tokens, or private participant information.

5. Functional test catalogue

All cases in this section are planned.

Test ID	Scenario and method	Expected result
TC-AUTH-01	Register with a unique email, valid password, and display name through the UI and API.	One account is created and the user is authenticated or directed to sign in.
TC-AUTH-02	Submit missing fields, malformed email, weak or invalid values, unexpected fields, and an already registered email.	Field-specific `4xx` errors are shown; no duplicate account is created.
TC-AUTH-03	Sign in with correct credentials, incorrect password, and an unknown email.	Correct credentials create a session; both invalid cases show the same non-disclosing error.
TC-AUTH-04	Sign out, then revisit a protected page and repeat a protected API request.	The session is invalidated and protected access requires authentication again.
TC-PRO-01	Edit the current user's profile with valid and invalid values, reopen it, then attempt to edit another profile through the API.	Valid values persist; invalid values are rejected; cross-user modification is denied.
TC-PRO-02	Open another student's profile and inspect the UI and API payload.	Only approved visible profile fields and current offered/wanted skills appear; private fields are absent.
TC-SKL-01	Add and remove an offered skill, attempt a duplicate, and inspect search/matching plus historical requests.	Active results change, duplicates are rejected, and historical records remain intact.
TC-SKL-02	Add and remove a wanted skill, attempt a duplicate, and inspect matching plus historical requests.	Active matches change, duplicates are rejected, and historical records remain intact.
TC-SKL-03	Select the same catalogue skill for two profiles and attempt to create an unsupported duplicate through the public API.	Both relationships use the same skill ID and catalogue duplication is denied.
TC-SRCH-01	Search by offered skill, apply a category filter, clear both, and inspect returned profile IDs.	Only eligible profiles appear, the current user is excluded, and clearing restores the documented default state.

Test ID	Scenario and method	Expected result
TC-SRCH-02	Run a zero-result query and use the recovery action.	No profile cards appear; a clear empty state and an action to broaden the search are available.
TC-MAT-01	Seed reciprocal, one-directional, self, and unrelated profiles; request suggestions; remove one side of the only reciprocal pair and recalculate.	Only reciprocal users are suggested; self is excluded; removing either direction removes the suggestion.
TC-MAT-02	Inspect every returned suggestion and compare its explanation with active profile data.	At least one current skill in each direction explains every suggestion.
TC-MAT-03	Open matching with no offered skill, no wanted skill, and a complete profile with no reciprocal match.	Missing-profile guidance is distinct from the no-match state; manual search remains available.
TC-REQ-01	Send a valid request, then attempt a self-request, inactive skill pair, and duplicate pending request.	One valid `Pending` request and history entry are created; invalid requests are rejected without extra records.
TC-REQ-02	View incoming and outgoing lists as both participants and as an unrelated user.	Each list contains only the authenticated user's requests with correct participant, skills, status, and creation time.
TC-REQ-03	As recipient, accept one pending request and decline another; repeat each action and try as sender or unrelated user.	Only the recipient can make one valid transition; repeated and unauthorized changes are rejected.
TC-REQ-04	As sender, cancel a pending request; repeat cancellation and try to cancel accepted, declined, and another user's request.	Only the sender can cancel once from `Pending`; all forbidden transitions are rejected.
TC-REQ-05	Create and transition requests, then inspect history as both participants and an unrelated user; attempt history modification.	Every change has an immutable actor, status, and timestamp record visible only to participants.
TC-AGR-01	For an accepted request, record topic, future date/time, each format, and appropriate location/link; repeat for non-accepted and unrelated requests and invalid fields.	Valid details are visible to both participants; status, ownership, required-field, and format-specific rules are enforced.
TC-AGR-02	Update an agreement as each participant and as an unrelated user.	Participant changes persist for both users and update actor/time; unrelated access is denied.
TC-AGR-03	Inspect navigation, UI, and OpenAPI operations for chat/video; record an in-person location and external meeting information.	No built-in chat or video feature exists; supported external/in-person details can be recorded.

6. Non-functional test catalogue

All cases in this section are planned.

Test ID	Method and conditions	Target / expected result
TC-SEC-01	Register a known password, inspect stored credential data, and verify correct/incorrect passwords through automated tests.	No plaintext or reversible password is stored; an adaptive salted hash is used; only the correct password verifies.
TC-SEC-02	Build an endpoint authorization matrix covering anonymous, owner/participant, authenticated non-participant, and wrong-role access for every protected operation.	Anonymous access returns `401`; unauthorized object access returns `403` or non-disclosing `404`; permitted access succeeds.
TC-SEC-03	Submit invalid types, lengths, formats, extra fields, path/query values, and common SQL-injection strings to every write/search endpoint.	Requests receive safe `4xx` responses; injection strings are treated as data; database structure and unrelated rows do not change.
TC-SEC-04	Inspect staging HTTP/HTTPS behaviour, production cookie attributes, repository history, and secret-scanning output.	HTTP redirects or is unavailable; cookies use `Secure`, `HttpOnly`, and explicit `SameSite`; no active secret is committed.
TC-SEC-05	Make 10 failed logins and then an additional attempt for the same account or client; repeat after the 15-minute window.	Attempts above the configured threshold return `429`; access becomes available after the window expires.

Test ID	Method and conditions	Target / expected result
TC-SEC-06	Compare profile/account API responses with approved schemas and inspect logs/errors for private fields.	Password hashes, sessions, secrets, internal configuration, and non-approved email data are absent.
TC-USA-01	Five representative students independently perform the full main scenario on staging using the protocol in Section 9.	At least 4 of 5 complete the scenario without moderator assistance.
TC-USA-02	Review registration, login, profile, search, matching, request, and agreement screens under normal, loading, success, validation, empty, and recoverable-error states.	Every applicable state provides visible and action-specific feedback; field errors identify the affected field.
TC-USA-03	Run automated WCAG checks on core screens; manually complete the main scenario by keyboard and inspect focus, labels, and contrast.	No critical automated violation; keyboard completion succeeds without a mouse; tested components meet WCAG 2.2 AA contrast.
TC-PERF-01	After warm-up, run three 10-minute tests with 20 concurrent simulated users against 1,000 users and 10,000 relationships; report the median run and every run.	In every run, p95 is at most 500 ms for profile/request operations and 1,000 ms for search/matching; API error rate is below 1%.
TC-COM-01	Execute the main scenario in the latest two stable major versions available at test time of Chrome, Firefox, Edge, and Safari; record exact versions.	Every browser/version combination passes or any blocking difference is fixed and retested before release.
TC-COM-02	Manually check all core screens at 360, 768, 1024, and 1440 px widths.	No unintended horizontal scrolling, clipped content, or hidden primary action.
TC-REL-01	Induce a failure between request-status update and history insertion in an integration test, then repeat without failure.	Failure rolls back both changes; success commits both atomically.
TC-MNT-01	Validate the versioned OpenAPI file and run contract tests for every main-scenario endpoint.	The document is valid and implemented schemas, authentication, responses, and error codes match it.
TC-MNT-02	Review the release traceability table and attached evidence.	Every Must requirement has at least one completed evidence record and no evidence field is empty.

`NFR-PERF-02` is a **Should** requirement. If the staging UI is available, Lighthouse mobile tests will also measure Largest Contentful Paint on every core page, with a target of at most 2.5 seconds. This optional check does not replace `TC-PERF-01`.

7. Mandatory-requirement traceability

The following table covers every requirement marked **Must** in SSE-6. Evidence links are intentionally recorded as planned test IDs until execution; they shall be replaced or supplemented with report links at release review.

Requirement	Planned verification evidence
FR-AUTH-01	TC-AUTH-01, TC-AUTH-02
FR-AUTH-02	TC-AUTH-03
FR-AUTH-03	TC-AUTH-04
FR-PRO-01	TC-PRO-01, TC-SEC-02
FR-PRO-02	TC-PRO-02, TC-SEC-06
FR-SKL-01	TC-SKL-01
FR-SKL-02	TC-SKL-02
FR-SKL-03	TC-SKL-03
FR-SRCH-01	TC-SRCH-01

Requirement	Planned verification evidence
FR-SRCH-02	TC-SRCH-02, TC-USA-02
FR-MAT-01	TC-MAT-01
FR-MAT-02	TC-MAT-02
FR-MAT-03	TC-MAT-03, TC-USA-02
FR-REQ-01	TC-REQ-01
FR-REQ-02	TC-REQ-02, TC-SEC-02
FR-REQ-03	TC-REQ-03
FR-REQ-04	TC-REQ-04
FR-REQ-05	TC-REQ-05, TC-REL-01
FR-AGR-01	TC-AGR-01
FR-AGR-02	TC-AGR-02
FR-AGR-03	TC-AGR-03
NFR-SEC-01	TC-SEC-01
NFR-SEC-02	TC-SEC-02
NFR-SEC-03	TC-SEC-03
NFR-SEC-04	TC-SEC-04
NFR-SEC-05	TC-SEC-05
NFR-SEC-06	TC-SEC-06
NFR-USA-01	TC-USA-01
NFR-USA-02	TC-USA-02
NFR-USA-03	TC-USA-03
NFR-PERF-01	TC-PERF-01
NFR-COM-01	TC-COM-01
NFR-COM-02	TC-COM-02
NFR-REL-01	TC-REL-01
NFR-MNT-01	TC-MNT-01
NFR-MNT-02	TC-MNT-02

8. Request-state verification matrix

Initial state	Actor and action	Expected state/result	Test case
No request	User sends a valid request to another user	`Pending`; initial history entry created	TC-REQ-01
No request	User sends request to self or with invalid skill pair	Rejected; no request/history created	TC-REQ-01

Initial state	Actor and action	Expected state/result	Test case
Pending	Recipient accepts	`Accepted`; one history entry created	TC-REQ-03, TC-REQ-05
Pending	Recipient declines	`Declined`; one history entry created	TC-REQ-03, TC-REQ-05
Pending	Sender cancels	`Cancelled`; one history entry created	TC-REQ-04, TC-REQ-05
Pending	Wrong actor accepts, declines, or cancels	Rejected; no data change	TC-REQ-03, TC-REQ-04
Accepted	Either participant records/updates agreement	Status remains `Accepted`; agreement is stored	TC-AGR-01, TC-AGR-02
Accepted, Declined, or Cancelled	Actor attempts a terminal transition	Rejected; no data change	TC-REQ-03, TC-REQ-04

9. Evaluation of project aim

9.1 Evaluation questions

The evaluation will answer:

1. Can representative students complete the full journey without moderator assistance?
2. Does reciprocal matching help students identify a suitable partner more efficiently than basic desired-skill search?
3. Do students understand why a suggested partner matches them?
4. Do users consider the proposed request and agreement workflow clear enough to use?

9.2 Participants and ethics

Recruit five university students who were not involved in implementing the tested screens. Participation shall be voluntary. Before testing, explain the purpose, activities, duration, data recorded, withdrawal option, and anonymisation approach. Use participant codes `P01`-`P05`; do not collect passwords, private messages, or unnecessary identifying data in repository evidence.

9.3 Main-scenario usability test

Each participant will use a fresh test account and receive a short goal without step-by-step interface instructions:

> Create an account, add the skills you can offer and want to learn, find a suitable partner, send an exchange request, respond as the second participant, and record the agreed topic, time, and format.

The moderator may clarify the task goal but shall not identify controls or give navigation instructions. Assistance requested or provided is recorded. A scenario counts as successful only when all required records are stored correctly and the participant completes it without moderator assistance.

9.4 Matching-versus-baseline comparison

The baseline is **manual search by desired skill without reciprocal-match suggestions**. The comparison condition uses reciprocal suggestions with an explanation of the skill match.

Each participant completes two equivalent partner-finding tasks on the same staging build and equally sized synthetic datasets:

- **Baseline condition:** search for profiles offering the desired skill, inspect profiles, and select a partner manually.
- **Matching condition:** inspect reciprocal suggestions and select a partner using the displayed match explanation.

Task order is counterbalanced: three participants start with the baseline and two start with matching. The datasets use different names but equivalent numbers of profiles, relevant offers, distractors, and one clearly suitable reciprocal partner. No participant sees the same dataset twice.

9.5 Metrics and targets

Metric	Measurement	Target
Full-scenario completion	Participants completing all main-scenario steps without moderator assistance	At least 4 of 5 (80%)
Partner-selection success	Participant selects the profile satisfying both directions of the assigned skill need	At least 4 of 5 in the matching condition
Matching explanation comprehension	Participant correctly states what each person can teach the other after selection	At least 4 of 5
Discovery time	Time from presentation of the partner-finding task to selection	Matching-condition median no longer than baseline median; a 20% reduction is the desired improvement
Discovery effort	Number of result/profile views before selection	Matching-condition median no higher than baseline median
Critical errors	Actions that prevent completion or create an invalid request/agreement	Zero unresolved critical errors at release
Task ease	One 1-7 rating after each major task, where 7 is very easy	Median at least 5 for the main scenario
Overall usefulness	1-5 agreement with “This platform would help me arrange a skill exchange”	Median at least 4
Matching usefulness	1-5 agreement with “The suggestions helped me find a suitable partner”	Median at least 4

The 20% discovery-time reduction is a desired result, not a release gate. With five participants, results will be reported descriptively using counts, medians, ranges, and anonymised observations; no population-wide statistical claim will be made.

9.6 Feedback collection

The moderator records task completion, time, assistance, errors, route taken, and short think-aloud observations. After each major task, the participant gives the 1-7 ease rating. After both discovery conditions and the complete scenario, ask:

- What made it easy or difficult to find a partner?
- Did the match explanation make sense? What was unclear?
- Which profile or search information was missing?
- Was sending and responding to the request clear?
- Were the agreed topic, time, and format fields sufficient?

- What would prevent you from using this platform?

Closed ratings and open comments shall be reported separately. Negative and contradictory feedback must be retained in the summary rather than removed as outliers.

10. Entry, suspension, and exit criteria

10.1 Entry criteria

- SSE-6 requirements and SSE-7 architecture are merged into `main`.
- The tested feature has acceptance criteria implemented and code-reviewed.
- Automated unit/integration checks for the feature pass.
- Staging is reachable over HTTPS and uses the documented test dataset.
- OpenAPI reflects the tested endpoints.
- Usability participants have received the consent information and test accounts.

10.2 Suspension criteria

Testing pauses when the environment loses data integrity, the build cannot complete the main scenario, a security issue could expose participant data, or a blocking defect makes subsequent results invalid. The reason and affected cases shall be recorded before rescheduling.

10.3 Release exit criteria

- every Must requirement has completed evidence in the traceability table;
- no unresolved critical or high-severity defect blocks the main scenario, authorization, privacy, or data consistency;
- security, performance, compatibility, responsive, accessibility, and OpenAPI checks meet their mandatory targets or have an explicitly approved corrective plan;
- at least 4 of 5 participants complete the full scenario without assistance;
- failed cases are fixed and retested, or formally recorded as accepted limitations by both team members and the supervisor;
- the final evaluation report separates measured results from interpretation.

11. Responsibilities and planned outputs

Activity	Responsible	Reviewer / participant	Output
Maintain requirement traceability and API/security tests	Nursultan Makhmutov	Tamir Burlibaev	Automated reports and traceability updates
Maintain UI, accessibility, responsive, and browser checks	Tamir Burlibaev	Nursultan Makhmutov	UI checklists and compatibility matrix

Activity	Responsible	Reviewer / participant	Output
Prepare staging data and load-test dataset	Nursultan Makhmutov	Tamir Burlibaev	Seed version and environment record
Moderate usability sessions	One team member	Other team member observes and records	Anonymised observation sheets
Analyse results and agree corrective actions	Both team members	Supervisor where needed	Evaluation summary and issue links

12. Review and delivery checklist

- [x] Every Must requirement is linked to at least one planned test or review.
- [x] Successful flows, invalid input, access permissions, and request status changes are covered.
- [x] Metrics, targets, measurement conditions, and required data are defined.
- [x] A manual-search baseline is defined for comparison with reciprocal matching.
- [x] User feedback and task completion assessment are defined.
- [x] Planned tests are distinguished from completed tests.
- [ ] Tamir has reviewed this plan through a pull request.
- [ ] Review comments have been resolved and the changes merged into `main`.
- [ ] The pull request is linked to SSE-9.

Suggested branch: `SSE-9-verification-evaluation-plan`.

10. Project management, responsibilities, and risks

Source document: docs/planning/project-plan-and-risks.md

Project Plan and Risk Register

Project: Development of a Peer-to-Peer Skill Exchange Platform for University Students

Team: Nursultan Makhmutov and Tamir Burlibaev, SE-2413, Astana IT University

Supervisor: Konstantin Latuta

Document owner / SSE-10 assignee: Tamir Burlibaev

Required reviewer: Nursultan Makhmutov

Updated: 23 October 2026

Status: Ready for pull-request review; review and merge evidence remain to be recorded.

Time zone: Astana, UTC+5, for all deadlines and targets.

1. Purpose, scope, and project links

This plan assigns responsibilities, sequences the work, defines milestone exits, and records risks and controls. It covers both the first assessment checkpoint and subsequent application development.

Resource Link or repository path

GitHub repository HashiraOfSun/student-skill-exchange

YouTrack board Student Skill Exchange Kanban

Planning task SSE-10 - Document project milestones, responsibilities, and risks

This deliverable `docs/planning/project-plan-and-risks.md`

Scope Project proposal

Requirements Initial Platform Requirements supplied for SSE-6

User evidence Survey report, SSE-2

Existing systems Comparison, SSE-4

Interface Prototype walkthrough, SSE-8

The first version includes accounts, editable profiles, offered and wanted skills from a shared catalogue, skill/category search, explained reciprocal recommendations, exchange requests and status history, and recording a shared topic, future time, and meeting format. Sessions take place in person or through an external service.

Request states are Pending, Accepted, Declined, and Cancelled. A pending request can be accepted or declined by its recipient or cancelled by its sender. Participants can edit an accepted exchange's shared plan without changing that status. The UI, API, persistence, and tests must implement the same rules.

Payments, built-in chat/video, native mobile applications, machine-learning recommendations, certificates, formal skill assessment, and group classes are outside this version.

2. Responsibilities and actual task allocation

Team responsibilities

Area Responsible member Required contribution and handoff

Backend and data Nursultan Makhmutov Architecture, schema/migrations, REST API, authentication, authorization, validation, matching, request processing/history, agreement storage, and backend documentation

Frontend and UI/UX Tamir Burlibaev User flows, prototype, responsive screens, forms, search/match explanations, request and agreement interfaces, loading/empty/error states, accessibility, and API integration

Research and requirements Both, according to the task register Nursultan leads literature, technical choices, and requirements; Tamir leads student research and system comparison; each reviews the other's conclusions

Integration and verification Nursultan: API/security/data; Tamir: client/usability Agree the contract, integrate increments, test core and negative scenarios, and link results to requirement IDs

Planning and reporting Tamir coordinates planning and user documentation; Nursultan coordinates technical evidence and assessment assembly Maintain risks, contribution evidence, setup/API/database documentation, report, and demonstration materials

Each work item has one assignee and the other member as peer reviewer. Supervisor feedback is recorded separately from peer approval.

Task register - supplied YouTrack snapshot, 23 October 2026

The displayed assignees and priorities below are transcribed from the supplied task list. Normal corresponds to “Обычная”; Major corresponds to “Серьезная”. For the technical workstream, this plan assigns person-level delivery responsibility to Nursultan while preserving the displayed account label `admin`. The screenshot does not establish that account's personal identity; record its owner in the contributor information before attributing contributions to an individual.

Task Title Displayed assignee Priority Delivery responsibility under this plan

SSE-1 Prepare the project proposal and define its scope `admin` Normal Nursultan

SSE-2 Interview students about skill exchange Tamir Burlibaev Normal Tamir

SSE-3 Review literature on peer learning and skill matching `admin` Normal Nursultan

SSE-4 Compare existing skill exchange and peer learning platforms Tamir Burlibaev Normal Tamir

SSE-5 Compare technology options and justify the selected stack `admin` Normal Nursultan

SSE-6 Define functional and non-functional requirements `admin` Major Nursultan

SSE-7 Design the system architecture and create diagrams `admin` Normal Nursultan

SSE-8 Create a UI prototype for the main user scenarios Tamir Burlibaev Normal Tamir

SSE-9 Prepare the verification and evaluation plan `admin` Normal Nursultan

SSE-10 Document project milestones, responsibilities, and risks Tamir Burlibaev Normal Tamir

SSE-11 Obtain supervisor feedback and document revisions `admin` Major Nursultan; Tamir addresses comments in his areas

SSE-12 Prepare and submit the assessment materials `admin` Major Nursultan coordinates; both verify their deliverables

Priority does not replace dependency order or a confirmed deadline. SSE-6, SSE-11, and SSE-12 need particular attention because they connect the evidence, decisions, and submission package. Task completion must be verified through deliverables and PRs; the screenshot alone is not review or merge evidence.

3. Current baseline and confirmed deadlines

The proposal and detailed initial requirements have been supplied. A six-response survey report, existing-system comparison, and standalone clickable prototype with a walkthrough are prepared for review. Architecture, technology, verification, supervision, and submission evidence must be checked in their respective tickets before the assessment package is declared complete.

The prototype demonstrates local interactions with sample data. Backend, database, integration, security, and multi-user readiness require separate implementation evidence.

Item Confirmed deadline Source / qualification

SSE-2 23 October 2026, 16:00 Supplied research task description

SSE-4 23 October 2026, 20:00 Supplied comparison task description

SSE-10 23 October 2026, 21:00 Current planning task description

SSE-8 24 October 2026, 10:00 Supplied prototype task description

First assessment checkpoint 24 October 2026; exact submission hour not established Supplied assessment instructions; confirm the cutoff through the assessment channel

Other tasks and later academic submissions/defence Not established by the supplied evidence Preserve any deadlines already recorded in live tickets; add missing dates with their source

A prepared file is not evidence that its task met its deadline. Use actual acceptance, review, merge, and submission timestamps. If a deadline passes with work outstanding, record the remaining work and revised decision while retaining the original date.

First-checkpoint exit gate

The checkpoint requires analysis, initial design, planning, supervision, and traceable contributions; a complete application is not required. SSE-12 can proceed when the following are checked:

Proposal, scope, project README, links, roles, and access for the team, supervisor, and committee.

Actual relevance evidence, with findings separated from assumptions. Record acceptance of the six-response questionnaire alternative to SSE-2's original 3-5-discussion brief.

Literature review with at least 12 meaningfully used sources, including eight scientific publications; comparison of 3-5 systems using 6-10 consistent criteria; at least three technology decisions with 2-3 alternatives each.

Prioritized requirements and verifiable criteria; consistent context, use-case, and draft component diagrams with editable sources.

Accessible prototype and walkthrough; verification/evaluation plan with metrics, conditions, and a baseline; this plan and risk register.

Substantive supervisor feedback, revision decisions, reviewed changes, and individual task/commit/PR evidence.

4. Proposed development milestones

The table is a proposed sequence after the first checkpoint, not an official calendar. Week 1 means the first agreed development week after the checkpoint; subsequent weeks are relative to that start. Calendar dates and available hours must be agreed once later academic deadlines are known. No weekly-hour commitment or December submission date is assumed.

M1-M7 are milestone identifiers, not YouTrack task IDs. Create real implementation tickets with owners, estimates, and reviewed targets before starting their work.

Milestone Proposed window Lead Prerequisites Exit criteria

M1 - Research and requirements refinement Week 1 Tamir: evidence; Nursultan: requirements Checkpoint materials and feedback; SSE-2/3/4/6 Method decision and limitations recorded; findings reflected in priorities; remaining assumptions have owners and validation actions

M2 - Design and API agreement Week 2 Nursultan: architecture/API; Tamir: UI M1; SSE-5/7/8/9 Data model, diagrams, user flow, request rules, versioned OpenAPI contract, examples, errors, and timezone handling reviewed together

M3 - First connected implementation Week 3 Both, by component M2 contract and setup Registration/sign-in and profile editing work through the actual frontend, API, and database; persisted changes survive reload; basic ownership/error paths checked

M4 - Core feature implementation Weeks 4-5 Both, by component M3 and feature acceptance criteria Skills, search, explained reciprocal matching, requests/history, and accepted-plan recording delivered in reviewed increments with negative-case checks

M5 - Full integration candidate Week 6 Nursultan coordinates; Tamir integrates client Reviewed M4 increments, migrations, configuration Complete main journey works for two users; persistence and contract checks pass; setup and candidate version are reproducible

M6 - Testing and evaluation Week 7 Nursultan: API/security/matching; Tamir: UI/usability M5 and reviewed SSE-9 protocol Every Must requirement has evidence; permission/state/transaction checks pass; responsive/browser checks, user results, matching-baseline comparison, and limitations recorded

M7 - Final documentation and demonstration Week 8 Tamir coordinates; Nursultan supplies technical sections M6 results and reviewed corrections Report, diagrams, API/database/setup/test/user guides, contribution evidence, and demo script match the selected version

Contingency Proposed reserve after M7, sized during scheduling Both Outstanding defects or evaluation gaps Resolve blockers and repeat affected checks; prepare a new reviewed candidate where needed

Tests, integration, and documentation accompany each increment. M5-M7 consolidate evidence. Review capacity weekly, keep one primary active task per member, reserve time for review and corrections, and reduce optional scope before compromising mandatory checks.

Immediate checkpoint sequence

Tamir opens the SSE-10 PR; Nursultan reviews; Tamir addresses comments, obtains approval, merges, and links the PR to SSE-10. Keep the 21:00 deadline visible. In parallel, verify remaining checkpoint evidence, resolve SSE-11 feedback, and assemble SSE-12. If a required review or deliverable remains unavailable, record the blocker and actual readiness rather than treating the plan as completion of that work.

5. Dependencies and handoffs

Dependency Required handoff / reason

SSE-1 → SSE-2/3/4/5 Scope, target users, exclusions, and investigation questions guide research and alternatives

SSE-2/3/4 + SSE-5 → SSE-6 refinement Evidence, limitations, and technical decisions inform requirements; unresolved assumptions remain labelled

SSE-6 → SSE-7/8/9 Architecture, UI, and verification use the same requirement IDs, scenarios, states, and acceptance criteria; these tasks may run in parallel

Data model + API contract → parallel frontend/backend work Agree skill IDs, schemas, permissions, errors, statuses, and timestamps before dependent integration

Authentication/profiles → skills and exchange ownership Requests require real identities, persisted profiles, and server-side access rules

Offered/wanted skills → search → reciprocal matching Discovery uses canonical active skills; recommendations require both overlap directions and self exclusion

Identity + skills + state rules → requests/history Validate participants, skill pair, duplicates, and atomic state/history updates

Accepted request → shared plan Only accepted participants can record/edit a topic, future time, and format

Reviewed increments → M5 → M6 → M7 Evaluation requires a connected, reproducible candidate; final claims require measured results and recorded limitations

Initial deliverables + SSE-10 + SSE-11 revisions → SSE-12 Submission needs reviewed materials, feedback decisions, access, contribution links, and the exact version

Supervisor feedback can start on ready materials. Frontend fixtures can support parallel work, but mock-only functionality does not close integration tasks. Record blockers in YouTrack with the prerequisite issue, missing handoff, owner, and next action.

The checkpoint readiness path is requirements consistency, architecture/prototype/verification evidence, feedback and revisions, reviewed package, and tagged submission. Planning runs alongside research and design rather than blocking their start.

6. YouTrack, branches, review, and completion

Task and board rules

Every task records its actual YouTrack ID, one assignee, reviewer, output, acceptance criteria, priority, dependencies, and a confirmed deadline or proposed target. Milestone/risk labels in this document do not substitute for issue IDs.

Use the project's board stages as applicable:

Stage Meaning

Очередь - Queue Scoped and assigned, awaiting work

Разработка - Development Active research, documentation, design, or implementation

Ревью - Review PR opened, author checks recorded, peer review requested

Тест - Test Applicable verification and acceptance checks being completed

На тестовом окружении - Test environment Applicable runnable changes checked in the integration/staging environment

Готово - Done Acceptance, required review, checks, merge, and task links complete

Documentation-only tasks require content and consistency checks; they do not require application deployment. Record the applicable verification in the PR. A blocked task names the dependency and next action and stays visibly incomplete.

Required development workflow

Start from the reviewed `main` baseline. Use a task branch, for example `SSE-10-project-plan-and-risks`.

Make coherent commits using an identifiable contributor account. For this change use `SSE-10: document milestones, responsibilities, and risks`. Record research/design/documentation progressively as well as code.

Open a PR into `main`, for example `SSE-10: add project plan and risk register`. Link SSE-10, summarize the change, explain verification, and disclose gaps. Add the PR link to the task.

Nursultan reviews this document's allocation, timing, dependencies, risks, and completion rules. Tamir reviews Nursultan-authored changes. Resolve substantive comments in the branch and request re-review after material revisions.

Obtain approval of the final content, pass applicable repository checks, and merge using the configured method. Preserve the relationship between the PR and resulting commit.

Add the merge commit or permanent merged-file link and verification summary to YouTrack. Check any automatic integration before relying on it to close the issue.

Definition of done: the required deliverable exists at the correct path; acceptance criteria have evidence; the required peer review covers the final changes; applicable checks pass; changes are merged into `main`; and the PR/commit evidence is linked to the task. Mandatory defects remain blockers. Code additionally requires relevant component, contract, integration, and negative-case checks plus updated setup/documentation. Research documents require source, date, consistency, and evidence/assumption checks.

Submission version and access

The supplied first-submission form requires `prob-period-1`. Tag the exact reviewed commit and provide its direct GitHub tag link. The general instructions also list pre-defence tags; follow each submission's specific naming requirement. If two names are required for the same package, point both to the same commit and record the relationship.

Check that attachments and prototype materials match that version and that the team, supervisor, and committee can access them. Keep submitted tags fixed; record later corrections through the required procedure and a new reviewed version. The standalone prototype remains usable offline after download.

7. Risk register

Likelihood and impact are qualitative planning judgments. Low likelihood: possible without a specific warning; Medium: plausible and requiring monitoring; High: current conditions make disruption especially likely. Low impact: local correction; Medium: rework within a milestone; High: a confirmed deadline, mandatory outcome, data protection, or core flow is threatened.

Each risk has one accountable owner. R01-R15 are local risk IDs; controls are planned until their execution is evidenced in the relevant task.

ID Risk and trigger Likelihood Impact Mitigation and response Responsible person

R01 Checkpoint work or review misses a confirmed deadline; required evidence remains missing at the final check High High Inventory evidence, finish focused PRs, and prioritize review/corrections. Record outstanding work and a revised decision without replacing the original deadline Tamir Burlibaev

R02 Unknown official dates conflict with the proposed sequence; an announced cutoff precedes planned readiness Medium High Confirm the assessment hour and later dates; anchor milestone dates to available capacity and official cutoffs; record impacts and re-plan Tamir Burlibaev

R03 The survey alternative is not accepted or findings are overstated; six questionnaire records differ from the discussion brief Medium High Record the method/sample decision, preserve all six responses and limitations, and arrange permitted follow-up evidence if required; retain the distinction from interviews Tamir Burlibaev

R04 Reciprocal matching yields too few partners despite complete profiles Medium High Measure coverage on a defined dataset; retain manual search and clear no-match guidance; report sparse-match cases and review any proposed model change Tamir Burlibaev

R05 Optional requests displace mandatory functionality Medium High Maintain explicit exclusions and priorities; estimate changes and dependencies; defer optional scope or obtain a reviewed scope/date decision Tamir Burlibaev

R06 API/client drift produces incompatible fields, skill IDs, errors, or request states Medium High Review the versioned contract and fixtures together; integrate at M3 and after increments; update both components and affected checks before incompatible changes merge Nursultan Makhmutov

R07 A user accesses or changes another user's protected record Medium High Enforce backend object permissions, credential protection, and input validation; test unauthenticated/cross-user access; block release on exposure or unauthorized writes Nursultan Makhmutov

R08 Repeated/concurrent actions create conflicting outcomes or separate status and history updates Medium High Guard transitions and use atomic writes; test duplicates, simultaneous actions, and induced failures; correct inconsistencies and retest before release Nursultan Makhmutov

R09 Mobile, keyboard, or error handling prevents completion Medium High Check 360/768/1024/1440 px, focus/labels/keyboard use, state handling, and the agreed browser matrix; fix main-flow blockers and repeat affected checks Tamir Burlibaev

R10 Participants see different meeting instants or can save past/ambiguous times Medium High Agree UTC storage and explicit display timezone; validate future times consistently; test different browser timezones and date boundaries Tamir Burlibaev

R11 Absence, coursework, or an unfamiliar tool causes repeated estimate overruns Medium High Review capacity weekly, keep tasks small, time-box investigation, document handoffs, and reserve review/fix time; reassign feasible work and defer optional scope Tamir Burlibaev

R12 Hosting, migration, configuration, or data loss prevents reproducible execution Low High Maintain a local run path, example configuration, reviewed migrations, and a checked backup/restore procedure; verify both environments and retain a local demo fallback Nursultan Makhmutov

R13 Evaluation lacks participants, comparable baseline conditions, or a recorded tested version Medium High Agree SSE-9's protocol early, arrange participants, freeze version/data, and record failures and conditions; report evidence gaps and seek a revised evaluation decision Tamir Burlibaev

R14 Sources or comparison claims do not support requirements or conclusions Medium Medium Verify original/official sources and access dates, mark unverified features Unknown, and separate interpretation; revise unsupported claims before reuse Nursultan Makhmutov

R15 Account identity, missing review links, access, or mixed/tagged versions prevent contribution and submission traceability Medium High Identify contributor accounts, retain task/PR/revision evidence, check access and package consistency, and keep submitted tags fixed Nursultan Makhmutov

Review R01-R03 and R15 during each remaining checkpoint work session; review all risks weekly and at milestone exits. When a trigger occurs, create or update the actual YouTrack issue with the risk ID, owner, impact, next action, and response target. Escalate threats to a deadline, data protection, or core flow to the other member the same day; involve the supervisor for academic scope/date/evidence decisions.

8. Progress, evaluation, and change control

At an agreed weekly review, confirm capacity, inspect completed evidence and blockers, update risks, and select the next small tasks. Track accepted outputs, Must requirements with evidence, unresolved core-flow defects, and dependency/risk actions. Commit count alone is not a progress measure.

Changes to scope, the API, estimates, or proposed dates record their reason, affected requirement/task IDs, dependency impact, owner, reviewer, and decision. Only official instructions or an authorized recorded decision change an academic deadline.

SSE-9 defines environment, data, metrics, and baseline before evaluation. The initial SSE-6 usability target is four unaided successful completions among five representative students; it remains a target requiring review, not a result. A proposed matching baseline is one-direction skill search with the same users/catalogue; confirm it in SSE-9 and record reciprocal validity, coverage, and applicable discovery measures. Do not infer learning improvement from partner discovery alone.

Maintain requirements, diagrams, API documentation, and checks during development. Link every Must requirement to test/review evidence at release review. Prototype checks do not establish backend security, performance, transactions, or multi-browser implementation readiness.

9. SSE-10 acceptance and review record

The content covers assigned responsibilities, all required milestone stages, confirmed/proposed timing, dependencies, risk likelihood/impact/mitigation/ownership, project links, and the development workflow.

Remaining acceptance step Required evidence

Nursultan reviews the document Actual PR review, resolved comments, revision commits, and approval of the final content

Changes are merged into `main` Actual merge commit and permanent merged-file link

PR is linked to SSE-10 Actual PR URL and verification summary in the task before Done

These external steps are not yet recorded in this document. The planning PR and SSE-10 retain the authoritative completion evidence.

11. Supervisor feedback and revisions

Source document: docs/supervision/supervisor-feedback.md

Project: Development of a Peer-to-Peer Skill Exchange Platform for University Students

Task: SSE-11

Status: Supervisor feedback recorded; pending team review

Supervisor: Konstantin Latuta

Document owner: Nursultan Makhmutov

Team reviewer: Tamir Burlibaev

1. Review details

Field	Record
Review date	23 October 2026
Format	Verbal feedback
Participants	Konstantin Latuta and Nursultan Makhmutov
Evidence basis	Nursultan's record of the verbal conversation; no recording or verbatim transcript was made

The feedback below is a faithful paraphrase of the verbal conversation. It is not presented as a direct quotation.

2. Review scope and materials

The supervisor reviewed the project at a general level, including its topic, purpose, intended scope, and the team's plan to prepare the required written materials. The current repository and its project documentation were available as the working record:

- Project proposal
- Literature review
- Existing systems comparison
- Technology comparison
- Initial platform requirements
- System architecture and diagrams

- Verification and evaluation plan

Repository: <https://github.com/HashiraOfSun/student-skill-exchange>

No detailed document-by-document comments or line-level corrections were provided during this review.

3. Feedback and action log

ID	Supervisor feedback	Team response or decision	Responsible person	Status	Evidence / next step
SF-01	The project was approved to continue.	Continue with the approved project topic, aim, and current scope.	Nursultan Makhmutov and Tamir Burlibaev	**No change required**	Approval is recorded in this document; continue work through the project backlog.
SF-02	The team should prepare and write the project materials independently.	The team will author the documentation, preserve sources and decisions in the repository, and use peer review for changes.	Nursultan Makhmutov; Tamir Burlibaev as reviewer	**In progress**	Continue repository-based documentation and link completed work through pull requests.

4. Revisions resulting from the review

No content revision was requested by the supervisor. Therefore, no project document was changed in response to a correction or objection.

This feedback record itself is the required documentation change: it records the approval, the instruction to work independently, the team's response, ownership, and status.

5. Open items and planned next steps

Item	Reason	Responsible person	Planned next step	Status
Complete the remaining assessment preparation	The supervisor delegated preparation of the materials to the team.	Nursultan Makhmutov and Tamir Burlibaev	Complete the remaining backlog items and prepare the assessment package under SSE-12.	Planned
Request clarification if a new academic or scope question arises	No detailed corrections were requested in this review.	Nursultan Makhmutov	Contact the supervisor with the specific question and append any new feedback to this record.	As needed

No unresolved supervisor objection or required correction is currently recorded.

6. Review outcome

- **Overall conclusion:** the supervisor approved the project to continue.
- **Required revisions:** none were specified.
- **Team responsibility:** the students are responsible for preparing and writing the project materials.

- **Further review:** no follow-up date was specified; clarification will be requested if needed.

7. Team verification and delivery checklist

- [x] The review date and review scope are recorded.
- [x] The actual verbal feedback is documented as a paraphrase without invented detail.
- [x] Each feedback item includes the team's response, responsible person, and status.
- [x] The absence of requested revisions is stated explicitly.
- [x] Planned next steps and responsibility are recorded.
- [x] Nursultan confirms that this record matches the verbal feedback he received.
- [ ] Tamir confirms that this record matches the feedback reported to the team.
- [ ] The document is reviewed through a pull request and merged into `main`.
- [ ] The pull request is linked to SSE-11.

Suggested branch: `SSE-11-supervisor-feedback`.

Submission readiness statement

This report covers the eleven required areas using the documentation currently merged into the repository. A separate evidence PDF contains readable captures of the YouTrack board and task records. Before official submission, the team must verify access permissions, update the repository README with final links, create the pre-defense-1 tag, complete the university form, and save the submission confirmation.

Status	Item
Complete	Proposal, research evidence, literature review, system comparison, technology comparison, requirements, architecture, prototype, evaluation plan, project plan/risks, and supervisor feedback are available.
Complete	Four previously open PRs (#8, #9, #10, #12) were merged into main before this report was assembled.
Required before submission	Confirm that the supervisor and committee can access the repository, PDFs, prototype artifact, and YouTrack evidence.
Required before submission	Clarify with the university whether GitHub accounts are accepted where the announcement requests GitLab accounts.
Required before submission	Create the pre-defense-1 repository tag only after the final documentation update is merged.
Required before submission	Submit one team form and save a screenshot or PDF of the confirmation.